

Material Dignity Infrastructure

Los Angeles Metropolitan Stabilization

A Street-to-Home Pipeline Analysis

Charles J. DiBella

Principal Systems Architect

Working Paper - May 2026

ABSTRACT

Chronic street homelessness in Los Angeles persists because the physiological prerequisites for maintaining housing are absent at the moment of placement. Finland reduced long-term homelessness by 68 percent through a national welfare infrastructure that provides those prerequisites before housing entry. The United States places individuals in housing while they remain in acute physiological crisis, producing sub-40 percent retention despite 24 billion dollars in California expenditure since 2019 (UCSF Benioff Homelessness and Housing Initiative, 2023); consequently, the Material Dignity Infrastructure closes this sequencing gap. Phase Zero deploys MHRC-licensed sub-16 unit clusters delivering metabolic, neurological, and nutritional stabilization under a DSM-5 protocol funded through Medi-Cal sub-acute billing at 50 percent federal participation before any permanent placement occurs. This physiological clearance sequence precedes five parallel operations: ACT field engagement over 12 to 24 months; CARE Court and Assisted Outpatient Treatment for compelled pathways; 2,000 pre-matched units at One California Plaza; riparian encampment abatement through field mapping and Clean Water Act proceedings; and Ontological Permanence Architecture sustaining residents post-placement. The fiscal model generates an annual Efficiency Surplus ranging from 33 to 82 million dollars per tower; simultaneously, eight binary verification metrics constitute the Singular Prototype Threshold. Every metric must pass before network expansion proceeds.

Document Metadata

Keywords

Material Dignity Infrastructure, adaptive reuse, homelessness stabilization, Measure Alpha, Los Angeles Metropolitan Stabilization, Housing First, Assertive Community Treatment, CARE Court, Dunbar Pod, ontological security, population taxonomy, pipeline engineering, implementation science

JEL Classification

- R31. Housing Supply and Markets
- R53. Public Facility Location Analysis, Public Investment, and Capital Stock
- H75. State and Local Government, Health, Education, Welfare
- I14. Health and Inequality
- I38. Government Policy, Provision and Effects of Welfare Programs
- K25. Real Estate Law

Target eJournals

- Housing and Residential Economics
- State and Local Government eJournal
- Public Health Law and Policy eJournal
- Property, Land Use, and Real Estate Law eJournal
- Urban Economics and Regional Studies eJournal
- Public Economics, State and Local Government Finance eJournal

Statement of Necessity

Los Angeles is experiencing a three-dimensional structural failure involving housing, public health, and fiscal policy. Despite 24 billion dollars in expenditure since 2019, 46,000 unsheltered individuals remain on the street; furthermore, mortality among the unhoused has accelerated to create a 30-year life expectancy gap. Current systems fail because they violate causal necessity by deploying housing before the physiological prerequisites for maintaining tenure are present. Whereas Finland solved chronic homelessness by providing these prerequisites through a comprehensive national welfare infrastructure, the United States lacks such a foundational layer. The Material Dignity Infrastructure engineers a substitute for this missing infrastructure: a Metabolic Stabilization layer as Phase Zero that restores biological prerequisites before any permanent unit is offered. Because an alternative framework built on this causal foundation is required, the Material Dignity Infrastructure proposes a falsifiable, industrially specified pipeline extending from street engagement through permanent residential stabilization.

Author Note

This framework is developed as an open-source collaborative infrastructure project. Source files, architectural models, and version history are maintained publicly at: https://github.com/material-dignity/working_papers.

Charles J. DiBella

ORCID: 0009-0003-7162-9615

Suggested Citation

DiBella, C.J. (Working Paper, May 2026). *Material Dignity Infrastructure: Los Angeles Metropolitan Stabilization*. Material Dignity Infrastructure Series. Available at SSRN: <https://ssrn.com/abstract=6579600>

Contents

1	The Los Angeles Inflection Point	5
1.1	The Five Simultaneous Operations	5
1.2	The Governing Principle	7
1.3	Pipeline A: The Near-Homeless and Voluntarily Transitioning	8
1.4	Pipeline B: The Encamped but Engageable	8
1.5	Pipeline C: The Behaviorally Calcified Chronic	8
1.6	Pipeline C Sub-Variant: The Hidden Riparian Population	9
1.7	Pipeline D: The Voluntary Nomadic	10
1.8	The Population Calculus	11
2	Phase Zero: Metabolic Stabilization	13
2.1	The Three-Gate Triage Decision Architecture	13
2.2	The Civil Liberties Architecture	14
2.3	The Three Concurrent Intake Pathways	14
2.4	The MHRC Cluster Model	15
2.5	The STC 65 Phase Zero Module	15
2.6	The Metabolic Stabilization Clinical Protocol	16
2.7	The Three-Phase Diagnostic Protocol	16
2.8	Day-30 Functional Indicators	17
3	Phase One: Pre-Admission Field Architecture	18
3.1	Stage 1: The By-Name HMIS Registry	18
3.2	Stage 2: ACT Team Sustained Presence	18
3.3	Stage 3: The Warm Offer Window	19
3.4	Stage 4: Encampment Resolution	19
3.5	ACT Workforce Sizing	20
3.6	Workforce Funding Stack	22
4	The Legal Lever System: Parallel Compelled Pathway	24
4.1	CARE Court (The CARE Act, 2023)	24
4.2	AOT: Assisted Outpatient Treatment	25
4.3	LPS Conservatorship (SB 43 Expansion)	25
4.4	The Three-Instrument Decision Matrix	27
5	Phase Two: Ground Floor Intake and Operational Continuity	28
5.1	The Open Resource Center	28
5.2	Zone A: The Sally Port	28
5.3	Zone B: Triage and Acuity Segmentation	29
5.4	The Incident Pathway and Outbreak Design	29
5.5	Acuity-Segmented Hygiene	30
5.6	Intake Stabilization Sequence	30
6	Phase Three: Ontological Architecture	31
6.1	The Acoustic Sanctuary: STC 65	31
6.2	The ALMU: Sovereign Cabin	31
6.3	Dunbar Verticality: Partitioning Against Anonymity	33
6.4	Biophilic Infrastructure: Two-Tier System	34
7	Social Infrastructure: The Human Layer	35
7.1	The Pod Steward: The Gardien Model	35
7.2	Resident Civic Engagement Pathway	35
7.3	The Pod Common Kitchen	36

7.4	Animal Companions: The On-Site Facility Model	36
7.5	The Residential Behavioral Floor	37
7.6	The Three Ps Infrastructure	37
7.7	Digital Access Center	38
7.8	Micro-Community Governance	38
8	The Cybernetic Grid: Automated-Human Operational Balance	39
8.1	The Telemetry Threshold	39
8.2	CPTED: Natural Surveillance Over Digital Surveillance	39
8.3	The Robotic Platform: The Gausium Scrubber 50	40
8.4	Nocturnal Access Protocol	40
8.5	Transportation Asset Management	40
8.6	Measure Alpha Cybernetic Allocation	41
9	Fiscal Architecture and the Efficiency Surplus	42
9.1	Layer 0: Medi-Cal Sub-Acute Billing Through MHRC Licensure	42
9.2	The Four-Layer Funding Stack	42
9.3	The Commercial Real Estate (CRE) Bailout Logic	43
9.4	Prototype Capitalization	43
9.5	The Efficiency Surplus and Operational Tolerance	44
9.6	Measure Alpha Integration	45
9.7	Non-Profit Co-Location Economics	45
9.8	Seven-Stakeholder Value Hierarchy	45
10	Legal Governance: The Stewardship Authority	46
10.1	The TVA Model: Operational Autonomy	46
10.2	The Home Rule Stabilization Ordinance	46
10.3	Sovereign Acquisition via Receivership	47
10.4	The Stewardship Contract	47
10.5	The Rapid Transition Protocol	48
10.6	Federal Supremacy Leasing	49
10.7	Section 1115 Demonstration Waiver	49
11	Comparative Models and Evidence Base	50
11.1	Houston: The Way Home Validation	50
11.2	Finland: National Housing First Program	50
11.3	The French Gardien Model	51
11.4	EmPATH: Emergency Psychiatric Assessment Treatment and Healing	51
11.5	CIT: Crisis Intervention Team Memphis Model	52
11.6	Adaptive Reuse Precedents	52
12	The Prototype Verification Threshold	53
12.1	The Eight Verification Metrics	53
12.2	The SPT Gate	54
12.3	The Falsifiability Clause	54
13	Conclusion	56
A	Operational Lexicon	58
	References	70

1. The Los Angeles Inflection Point

The chronic street homelessness crisis in Los Angeles persists because the metropolitan urban design has reached a point of structural exhaustion since the city was built upon a post-war ideal of low-density residential ownership and automobile dependency. Contemporary land values and infrastructure costs have rendered that model obsolete for the average worker whereas the resulting region is characterized by a profound housing imbalance. An abundance of per capita luxury inventory exists alongside a total absence of "utility housing" for the working class to create an architectural trap that forces a transportation tax on the poor while leaving the most vulnerable with no soft landing in the private real estate market. Although the city contains 50 million square feet of vacant commercial inventory, it has failed to adapt these assets for human stabilization (GlobeSt, 2025). Since fiscal year 2019, California has spent 24 billion dollars on homelessness programs while the encampment population continued to grow; the crisis is not a matter of capital volume but of engineering sequence (California State Auditor, 2026; Los Angeles Homeless Services Authority, 2025).

Legacy interventions fail because they violate the causal necessity of stabilization since municipal encampment clearances produce a "Leaf Blower Effect" that generates visible motion while achieving zero net population reduction. These operations lack a simultaneous destination whereas congregate shelter models generate rational refusal by forcing individuals to choose between residency and survival anchors such as animal companions, partners, or personal possessions (United States Department of Agriculture, 2023; U.S. Department of Housing and Urban Development, 2022). Fragmented clinical outreach fails to match the multi-year engagement window required by a neurologically compromised population so that these failures converge at a single point where the city offers placement to individuals who have not yet regained the physiological capacity for tenancy.

The Material Dignity Infrastructure (MDI) resequences this pipeline through three operational layers: a Clinical Field Architecture, a Legal Lever System for treatment-resistant cohorts, and a Terminal Infrastructure Node. Within this framework, the tower at One California Plaza functions as the stabilization destination that bridges the gap between the street and the sovereign real estate market. Because the street environment precludes recovery, the tower provides the metabolic gateway required to restore an individual's biological prerequisites for long-term residential tenure.

1.1. The Five Simultaneous Operations

The objective is documented, sustained elimination of visible encampment homelessness within the defined geographic zone. The prototype operates against a fixed denominator of approximately 2,500

unsheltered individuals within the Skid Row containment zone, achieving an 80 percent reduction via the 2,000-unit capacity at the 24-month prototype verification threshold, alongside full remediation of adjacent riparian encampments. Achieving metropolitan-scale impact requires expanding this prototype into a 20-tower network. Executing the initial prototype requires five operations running in parallel.

The Field Architecture deploys Assertive Community Treatment teams to every chronic encampment corridor in the target zone, builds a continuous dynamic by-name HMIS registry of every individual, and executes sustained presence engagement over a standardized 12 to 24 month period. Because the registry tracks real-time behavioral data throughout this engagement window, each named individual is mapped and matched to a specific unit, pod, and floor prior to the final warm offer. The physical unit activation occurs through a phased occupancy rollout synchronized to the 75-bed Phase Zero stabilization throughput rather than a single simultaneous event.

The Legal Lever System operates a parallel compelled clinical pathway. The MDI Stewardship Authority functions as a petitioning entity under CARE Court for individuals whose anosognosia or grave disability makes voluntary engagement clinically impossible. Assisted Outpatient Treatment mandates psychiatric compliance for the treatment-resistant. Lanterman-Petris-Short (LPS) Conservatorship provides the legal guardian mechanism for the profoundly gravely disabled. These instruments create a court-supervised pathway running in parallel to voluntary outreach.

The coordinated unit activation opens the first tower at One California Plaza with 2,000 units pre-matched to named individuals in the by-name registry. This activation transforms the offer from a generic referral into a specific, immediate, and irreversible commitment integrating a named room, a stored cart, a kenneled pet in the Level 2 facility, and a keycard.

The Environmental Compliance Enforcement Track addresses the hidden riparian sub-population occupying flood control channels, storm drains, and protected wetlands across the Los Angeles basin. These encampments produce documented federal regulatory violations including fecal coliform contamination exceeding Clean Water Act National Pollutant Discharge Elimination System (NPDES) permit standards, biological waste infiltration into protected habitat, and infrastructure obstruction within Army Corps of Engineers jurisdiction (United States Congress, 1972). Systematic field mapping of riparian corridors, using ground-based documentation teams and available aerial survey methods operating within established Fourth Amendment parameters, identifies and locates hidden encampments within the target zone. Environmental abatement proceedings trigger simultaneously

with the warm offer whereas the legal basis for clearance is environmental protection. The MDI placement inventory provides the simultaneous destination required for clearance to be durable.

The Ontological Permanence Architecture sustains the individual inside the tower. The Dunbar Pod structure, STC 65 acoustic sanctuaries, on-site animal accommodation, and the Pod Steward rebuild the neurological conditions for selfhood that chronic street life destroys including predictability, privacy, control, and recognized social membership (Giddens, 1991; Laing, 1960).

1.2. The Governing Principle

Sovereign space operates as the clinical precondition for stabilization since placement represents the foundational requirement for clinical improvement rather than a reward for recovery.

Finland proved this at the national scale demonstrating a 68 percent reduction in long-term homelessness between 2008 and 2022 (Y-Foundation, 2023). Houston proved it at the metropolitan scale achieving a 60 percent reduction between 2011 and 2020 alongside 90 percent two-year housing retention (Coalition for the Homeless of Houston/Harris County, 2024). The MDI thesis applies this principle at the architectural and legal scale required for Los Angeles.

The existing unmanaged street environment imposes a continuous biological failure condition upon the encamped population. Los Angeles County mortality data document the consequences (Los Angeles County Department of Public Health, 2026). The problem is un-engineered. The Material Dignity Infrastructure engineers it.

The chronic unsheltered population is not a monolithic actor. MDI interventions fail when they treat all unsheltered individuals as a single cohort requiring a single solution. Five structurally distinct sub-populations require five separate intervention architectures. Misallocating instrument to population produces waste at best, compounded harm at worst.

These five pipelines function as a dynamic state machine rather than a static categorization. Because behavioral stability and clinical acuity fluctuate, individuals may transition between states as their needs change; a Pipeline B resident may regress to a Pipeline C state during a substance-induced crisis or stabilize into a Pipeline A state following metabolic intervention. The MDI tracking architecture logs these transitions in the by-name registry to ensure that clinical instruments adjust automatically to the individual's current state. By anticipating these transitions, the framework prevents instrument mismatch and maintains systemic responsiveness throughout the engagement window.

1.3. Pipeline A: The Near-Homeless and Voluntarily Transitioning

Economic disruption, temporary crisis, or institutional discharge from hospital, prison, or foster care triggers entry for this cohort, representing approximately 12,246 individuals across the region who present free of major psychiatric comorbidity and behavioral calcification. Because social functioning remains intact, this population self-identifies as needing placement and engages voluntarily with the Coordinated Entry System whereas the MDI ALMU model resolves the transition with minimal friction. CES functions as the existing pipeline that MDI accelerates by creating surplus transitional inventory. Pipeline A is not the primary driver of street visibility or public safety concern because these individuals cycle through the system faster and rarely form long-term encampments.

1.4. Pipeline B: The Encamped but Engageable

This cohort comprises approximately 16,485 individuals characterized by long-term encampment with functional social bonds within the camp community. Psychiatric comorbidity may be present but the individual retains enough insight to evaluate offers rationally. Refusal targets congregate shelter models whereas the stabilization infrastructure eliminates the barriers driving rejection.

The refusal pattern is consistent across the research literature (United States Department of Agriculture, 2023; U.S. Department of Housing and Urban Development, 2022). Refusal anchors on forced separation from animal companions required for emotional regulation, mandatory separation from intimate partners, and the loss of accumulated belongings representing the individual's only remaining assets. Eliminating these barriers drives acceptance rates upward. Houston's "The Way Home" achieved 58 to 70 percent encampment-level acceptance rates when specific and immediate offers were made (Coalition for the Homeless of Houston/Harris County, 2024). The MDI architectural model is superior to what Houston offered. The engagement timeline spans weeks to months executing sustained presence outreach followed by a specific warm offer at an acute crisis window.

1.5. Pipeline C: The Behaviorally Calcified Chronic

This cohort, representing approximately 16,014 individuals, constitutes the core of the street crisis and the primary driver of public safety concern. Because these individuals present with a Long Duration of Untreated Psychosis ranging from 5 to 15 years, often complicated by active schizophrenia spectrum disorder and Diogenes syndrome overlay, the population requires a distinct intervention instrument at the neurological level.

Anosognosia operates as a neurological incapacity to perceive one's own psychiatric condition. The frontal lobe damage producing psychosis simultaneously destroys the self-monitoring faculty driving self-recognition. 50 to 80 percent of individuals with schizophrenia present with anosognosia. Outreach strategies premised on persuasion target a faculty that metabolic disease has already destroyed.

Duration of Untreated Psychosis produces measurable progressive brain changes with each untreated year including increasing cognitive rigidity, treatment resistance, and functional decline. A decade of street-level untreated psychosis is a different neurological substrate than six months. Standard clinical timescales do not apply.

Diogenes syndrome overlay presents as extreme self-neglect, domestic squalor, social withdrawal, and adamant refusal of help. The accumulated hoard is not random accumulation. It functions as what the MDI framework terms an Exoskeletal Identity Structure. This structure operates as the physical boundary of a self-constructed world that has replaced the social world that rejected or terrified the individual. Removing the hoard without replacing the identity structure triggers acute psychiatric destabilization. Intervention must be identity-preserving.

Metabolic collapse operates as a cognitive impairment independent of psychiatric diagnosis. The individual presenting at intake is not operating from a healthy neurological baseline. Electrolyte derangements from sustained dehydration and malnutrition impair executive function, impulse control, and emotional regulation. Volatility at intake derives from physiological distress compounding the psychiatric presentation.

The Pipeline C behavioral pattern features active hostility to outreach, conspiratorial framing of offers, repeated encampment re-establishment following displacement, and preference for outdoor environments where the paranoid threat-perception system has habituated. Unfamiliar enclosed spaces trigger acute agitation. The necessary intervention combines ACT sustained presence over 12 to 24 months and the legal lever system targeting individuals refusing voluntary engagement.

1.6. Pipeline C Sub-Variant: The Hidden Riparian Population

A sub-set of Pipeline C operates invisible to standard Point-in-Time counts and resists both voluntary outreach and legal lever engagement. This cohort remains mobile, terrain-adaptive, and encamped in environmentally sensitive or legally inaccessible sites including the LA River riparian corridor, Ballona Creek, Santa Ana River, Hansen Dam Recreation Area, storm drain infrastructure, and urban edge creek systems. Estimated at 20 to 30 percent above the PIT count's Pipeline C figure due to systematic undercount of hidden sites.

This population has built a calibrated survival architecture demanding territorial stability, mutual protection hierarchies, known supply chains, animal companions as primary attachment bonds, and ontological security derived from competence in terrain survival. The MDI voluntary offer competes against a system that already operates on its own terms. The voluntary acceptance rate for this sub-population is 15 to 30 percent globally across all evidence-based programs regardless of offer quality.

Riparian encampments produce documented federal regulatory violations that transform this from a social services problem into an environmental enforcement matter. Fecal coliform bacteria contamination of LA waterways at levels exceeding Clean Water Act NPDES permit standards (United States Congress, 1972). Hypodermic needle and pharmaceutical waste infiltration into riparian and wetland habitat. Biological waste and trash accumulation blocking flood control infrastructure within Army Corps jurisdiction. LARWQCB enforcement records document these violations. They provide a legal basis for abatement independent of housing law, rooted in state-level nuisance protocols triggered by Section 402 National Pollutant Discharge Elimination System (NPDES) enforcement liabilities (United States Congress, 1972). Precedent exists for environmental eviction under state and federal water board mandates where life-safety and severe contamination risks supersede standard habitability protections (Environmental Protection Agency, 2023). Following the Supreme Court reversal of *Martin v. Boise* via *Grants Pass v. Johnson*, municipalities maintain unrestricted clearance authority. The MDI architectural model requires simultaneous shelter placement strictly to ensure clearance durability rather than legal compliance.

1.7. Pipeline D: The Voluntary Nomadic

The voluntary nomadic sub-population, comprising approximately 2,355 individuals, is not metabolically destabilized in the Pipeline C clinical sense. This cohort has organized a functioning life around outdoor living, street community bonds, informal economy, and deliberate rejection of institutional structures based upon genuine preference and cultural identity. The street sub-culture this population inhabits is real. Its social bonds are real. The competence, freedom, and mutual protection it provides are real. Treating Pipeline D individuals as a variant of Pipeline C is a diagnostic error with severe operational consequences.

Interventions designed for acute metabolic destabilization produce rational refusal when applied to individuals whose primary condition is a preference for outdoor community life. Misdirecting

that refusal into the legal lever system wastes legal resources and damages the trust-based outreach relationships required for voluntary engagement over time.

The correct intervention instrument for Pipeline D is harm reduction without a placement requirement since ground floor open access provides wound care, nutrition, hygiene, needle exchange, veterinary services, and device charging without an intake requirement, a clinical agreement, or any precondition. These services reduce harm even when placement does not occur while no element of the Pipeline D routine intervention requires legal authority excepting universal emergency 5150 WIC medical holds. No element requires the individual to surrender outdoor life in exchange for care because the Anti-Detention Covenant protects this population. The MDI system offers care without compulsion so that Pipeline D individuals may self-select into the MS Unit or ALMU at any point they choose whereas the standing offer does not expire.

The bulk majority argument establishes why Pipeline D management does not require clearance because when the metabolically destabilized and acutely psychotic individuals comprising the Pipeline C population are substantially housed, what remains on the street is a qualitatively different population. This cohort presents as smaller in volume and less acutely destabilized whereas the behavioral contagion effect that amplifies crisis in dense encampment conditions diminishes. Harm reduction services operate more effectively on fewer individuals with less acute presentations so that community tolerance for the remaining voluntary nomadic population increases because the character of the street environment has changed.

1.8. The Population Calculus

The five pipeline classifications, matched intervention instruments, and primary MDI delivery nodes are summarized in Table 1. Each row represents a distinct decision architecture. The table is read as a triage reference: the population classification determines the instrument, the instrument determines the node, and the node determines the operational sequence.

Pipeline	Timeline	Intervention	MDI Node
A: Voluntary	Days to weeks	CES rapid placement	Ground Floor, ALMU
B: Engageable	Weeks to months	Warm offer at crisis window	Field Team, ALMU
C: Calcified	12–24 months + legal	ACT + CARE Court + AOT/LPS	Field, Legal, ALMU
C: Riparian	Abatement + offer	field mapping + environmental enforcement	LARWQCB, Warm Offer, ALMU
D: Voluntary Nomadic	Open-ended	Harm reduction, no pre-condition	Ground Floor open access only

Table 1. Population Pipeline Classification and Matched Intervention Architecture. Pipeline D receives harm reduction services without housing requirement. No legal instrument applies to Pipeline D.

Pipeline A approaches full voluntary engagement within a standard coordinated entry timeline. Pipeline B reaches 58 to 70 percent acceptance at the targeted warm offer. Pipeline C visible cohorts generate 15 to 30 percent voluntary acceptance, with the remainder requiring the legal lever pathway. Pipeline C hidden riparian populations demonstrate minimal voluntary engagement, establishing environmental enforcement as the primary access mechanism. Pipeline D receives services without placement pressure. Clearance of Pipelines A through C from the street environment changes the character of the remaining Pipeline D population sufficiently that harm reduction becomes the dominant management instrument county-wide.

Because the pipeline classification is complete, the instrument matched to each population now enters its operational specification.

2. Phase Zero: Metabolic Stabilization

While Finland demonstrated a 68 percent reduction in long-term homelessness, its Housing First model simultaneously achieves 80 to 90 percent two-year housing retention for those placed (Y-Foundation, 2023; FEANTSA, 2022). The same intervention, applied in the United States, produces retention below 40 percent (UCSF Benioff Homelessness and Housing Initiative, 2023).¹ The mechanism difference is not the housing itself but rather the physiological condition of the individual at the moment of placement since Finland’s national welfare infrastructure delivers the nutritional support, healthcare, and social services that stabilize individuals before they enter housing. Whereas the United States delivers a placement unit to an individual who remains in acute physiological crisis, that individual cannot sustain tenancy so that the resulting failure stems strictly from neurological and metabolic incapacity.

Metabolic Stabilization is the core physiological intervention that closes this gap. It operates within Phase Zero of the MDI pipeline: the clinical stabilization stage that runs concurrently with ground floor intake to execute physiological stabilization before the permanent warm housing offer occurs. Without it, the pipeline delivers housing to individuals who cannot hold housing. With it, the pipeline replicates the Finnish prerequisite layer from a Los Angeles County institutional foundation.

2.1. The Three-Gate Triage Decision Architecture

Mobile crisis outreach workers, ACT teams, and law enforcement assess each individual presenting in crisis. Three authorized outcomes exist for crisis presentations. The triage determination routes the individual to the pathway matched to their presenting condition.

Presenting Condition	Authorized Pathway	Governing Authority
Criminal act, danger to others, outstanding warrant	JAIL	Law enforcement, Courts
Acute medical crisis, overdose, severe injury, imminent psychiatric danger	EMERGENCY ROOM	Medical authority; 5150 Welfare and Institutions Code (WIC) if criteria met
Metabolic destabilization, psychiatric crisis without criminal act, intoxication, exposure	MS UNIT	mobile crisis outreach and ACT clinical determination

¹Community First! Village achieves 85 to 88 percent retention by operating a high-support community model distinct from standard Housing First placement (Mobile Loaves and Fishes, 2024).

Table 2. Three-Gate Triage Decision Architecture. Applied jointly by mobile crisis outreach, ACT team, and law enforcement at point of contact. Release to street is not an authorized triage outcome for individuals meeting crisis criteria. Pipeline D (Voluntary Nomadic) individuals are exempted from the triage gate unless independently meeting the strict 5150 medical hold standard. Non-crisis individuals retain full freedom of movement at all times.

This architecture does not create new coercive authority. California Welfare and Institutions Code Section 5150 already authorizes a 72-hour involuntary psychiatric hold for individuals posing danger to self, danger to others, or meeting grave disability criteria. This authority predates the Material Dignity Infrastructure whereas the mobile crisis outreach and ACT clinical determination identifies which existing legal pathway applies. MDI provides the destination those pathways currently lack.

2.2. The Civil Liberties Architecture

Existing California legal authority governs the acute triage window covering the hours immediately following mobile crisis, ACT, or law enforcement assessment for individuals in active crisis. This statutory authority includes medical holds under 5150 WIC, CARE Court mandates, AOT compliance enforcement, and LPS conservatorship pathways. This pre-existing statutory authority defines the boundary conditions of compulsion and forces individuals out of the street environment while the MDI residency itself remains voluntary. The Anti-Detention Covenant codified in the Stewardship Authority charter governs voluntary residency only, guaranteeing unrestricted egress for all residents at all times following admission, with a strict legal exception subordinating this right to active LPS conservatorship or CARE Court mandates. The Covenant governs what occurs after the acute crisis resolves and the individual is admitted as a participant rather than a detainee. Pipeline D individuals and any non-crisis individual who declines the Phase Zero offer retain full street access. The involuntary gate applies strictly to external court orders and acute crisis presentations, while the facility reserves the right to reject elopement-risk conservatees to maintain operational integrity.

2.3. The Three Concurrent Intake Pathways

Three distinct routes operate concurrently to deliver individuals into the Metabolic Stabilization sequence. Each operates independently. The intake architecture receives from all three simultaneously.

Pathway 1. Sally Port formal intake. This pathway relies on a mobile crisis outreach and ACT team-facilitated warm offer directed to a named individual on the by-name HMIS registry. The individual is pre-matched to a specific unit, pod, and floor before the offer conversation even begins. Because the process is structured, documented, and consent-based, the individual enters the system known, pre-matched, and at a chosen moment of willingness.

Pathway 2. Open ground floor walk-in. An individual presents voluntarily to the open-access resource center without prior registration or preconditions. Upon arrival, ground floor Zone B triage concurrently assesses the presenting condition and executes secure clinical transport to an external Metabolic Stabilization tier scattered site. Because walk-in access converts unknown individuals into registered participants, this serves as the highest-volume intake pathway during early operations.

Pathway 3. Secondary clinical transfer from field contact, law enforcement, or ER. Individual transferred following field contact, a non-criminal law enforcement determination, or ER discharge to a lower-acuity setting. The backdoor entrance to Zone A physically separates this pathway from the open resource center population, preventing crisis contagion in the commons and preserving the non-clinical character of open access.

2.4. The MHRC Cluster Model

Phase Zero operates through Mental Health Rehabilitation Center clusters deployed within the contiguous one-mile footprint of the Skid Row defined geographic zone (California Department of Health Care Services, 2024c). To process the 500 Pipeline C entries requiring 60 to 90 day stabilization windows without generating a backlog, the prototype mathematically requires the concurrent acquisition of 75 external MHRC beds spanning five localized clusters. Each cluster holds exactly 15 units. Each files an independent Department of Health Care Services (DHCS) Form 1813 MHRC licensure application. Each bills Medi-Cal independently as a specialty mental health sub-acute facility at approximately 50 percent federal financial participation through Federal Medical Assistance Percentage (FMAP). While the sub-sixteen unit threshold targets the Institution for Mental Disease Exclusion architecturally, the clustered design requires a Section 1115 demonstration waiver anchored to the California CalAIM framework to secure Centers for Medicare and Medicaid Services approval.

MHRC is a 24-hour DHCS-licensed facility classification for adults with mental disorders providing intensive rehabilitative services for sub-acute stabilization, skill-building, and independence. The 30 to 90 day Metabolic Stabilization protocol matches this classification precisely. Licensure requires submission of DHCS Form 1813 with a defined service array, staffing plan, and individualized treatment planning protocol (California Department of Health Care Services, 2024c,a).

2.5. The STC 65 Phase Zero Module

The physical module of each MHRC cluster is the STC 65 Phase Zero Module: a sixty-five-decibel-proof sound enclosure designated as a state-owned clinical environment. It is not a residential dwelling.

It is not a shelter. It is a therapeutic milieu unit analogous to a hospital step-down bed (Frontiers in Psychology Editorial Board, 2024; GB Acoustics, 2024).

Hypervigilance is a Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) diagnostic criterion for Post-Traumatic Stress Disorder. Auditory hypersensitivity is a documented symptom of methamphetamine-induced psychosis during post-acute withdrawal. Sleep continuity disruption from environmental noise prolongs both conditions by preventing the neurological consolidation processes occurring during slow-wave and rapid eye movement (REM) sleep. The sixty-five-decibel acoustic specification is a therapeutic milieu standard required to deliver effective psychiatric rehabilitation to a population whose primary diagnoses include conditions for which uncontrolled auditory stimulation is a documented symptom trigger and recovery barrier. This is the Form 1813 clinical language that defends the specification against regulatory challenge. The STC 65 classification as a medical facility exits the residential zoning framework that community opposition targets and enters the medical facility regulatory pathway where opposition carries materially different legal weight.

2.6. The Metabolic Stabilization Clinical Protocol

The Pipeline C methamphetamine-dependent population requires a specific clinical content sequence (Mehanna et al., 2008). Methamphetamine withdrawal produces psychosis, severe depression, and cognitive impairment persisting weeks to months after cessation. These are neurological states, not preference states. They prevent participation in any structured program until they resolve. The MHRC cluster provides the clinical environment in which neurological recovery is physiologically possible.

Four clinical targets govern the stabilization sequence. Nutritional refeeding and electrolyte correction addresses Thiamine deficiency producing Wernicke encephalopathy risk, Magnesium deficiency producing anxiety and seizure risk, and Protein deficiency producing impaired neurotransmitter synthesis. Sleep continuity restoration targets six or more hours of uninterrupted sleep without pharmacological support. Wound care and infection treatment resolves acute infectious disease states interrupting physiological recovery. Psychiatric symptom stabilization as withdrawal clears enables progressive diagnostic clarification from provisional to finalized DSM-5 diagnosis.

2.7. The Three-Phase Diagnostic Protocol

A Pipeline C resident arriving post-acute from the emergency room presents with no prior treatment history, significant cognitive impairment from methamphetamine withdrawal, and active psychosis

potentially indistinguishable from primary psychotic disorder. Billing requires a clinical reasoning trail. The protocol provides it in three phases (California Department of Health Care Services, 2024c).

Phase 1 (Hours 0–72): A Licensed Practitioner of the Healing Arts conducts clinical interview and mental status examination. Staff document observable symptoms including psychosis, cognitive impairment, withdrawal signs, and functional deficits. A provisional DSM-5 diagnosis is recorded: Stimulant-Induced Psychotic Disorder or Unspecified Schizophrenia Spectrum Disorder as appropriate. DSM-5 allows provisional diagnosis when full criteria are unclear due to substance effects. The assessment itself is billable as Mental Health Services Assessment under DHCS Specialty Mental Health Services rules.

Phase 2 (Days 1 to 14): Collateral from emergency room records, field notes, and by-name HMIS registry. Toxicology and basic labs to rule out acute medical issues. Daily progress notes tracking symptom changes as withdrawal clears. The daily note entry sustains provisional billing and constitutes the audit-defensible clinical reasoning trail.

Phase 3 (By Day 30): DSM-5 diagnosis finalized and signed by the LPHA. The Individualized Treatment Plan links problems to measurable goals and rehabilitative services. Full Medi-Cal billing for specialty mental health services is authorized following signed ITP.

2.8. Day-30 Functional Indicators

The ITP measurable goals define five functional indicators. Progress on at least two by day 30 constitutes the interim stabilization checkpoint, derived from standard psychiatric sub-acute utilization review criteria (California Department of Health Care Services, 2024c). Final Phase Zero clearance requires maintaining at least two indicators across the 60 to 90 day window. If fewer than two are met at the checkpoint, the ITP extends the stabilization phase with revised goals. Billing continues. The timeline adjusts to the individual.

1. Sustained hygiene maintenance without staff prompting
2. Sleep continuity of six or more hours without pharmacological support
3. Orientation to person, place, and date consistently across three consecutive days
4. Participation in at least one structured group activity per day
5. Initiation of at least one activity of daily living task independently per day

These indicators are the measurement substrate for Verification Metric 7. The Phase Zero clearance rate is the causal test of the primary MDI thesis. This rate represents the proportion of residents

meeting two or more indicators at 90 days. A retention gain at Metric 2 (housing retention) without a clearance gain at Metric 7 cannot be attributed to the Metabolic Stabilization sequencing design.

3. Phase One: Pre-Admission Field Architecture

The MDI tower is a terminal infrastructure node. It cannot function as an intake mechanism for Pipeline C without a pre-admission field architecture operating at the encampment level 12 to 24 months in advance. The Field Architecture is the pipeline. Without it, the tower houses Pipelines A and B while Pipeline C remains on the street, congregating at the base of the building.

3.1. Stage 1: The By-Name HMIS Registry

Field documentation teams deploy to all encampment sites within the MDI target zone during months one through six of pre-operations. Field documentation requires mapping each individual's known name or identifier, fixed versus nomadic GPS location pattern, observable survival anchors including animal companions and primary possessions, tentative Pipeline classification, and behavioral observation notes. All data enters an HMIS instance managed by the MDI Stewardship Authority. The by-name list is the operational backbone. Every housing unit pre-opening is pre-matched to a named individual on this list. The warm offer is specific because the preparation was specific.

Houston's HMIS integration across over 100 partner agencies was the single factor most cited in that city's 60 percent reduction (Coalition for the Homeless of Houston/Harris County, 2024). Real-time data eliminated the coordination lag where individuals fell through agency handoffs.

3.2. Stage 2: ACT Team Sustained Presence

Each encampment cluster receives a dedicated Assertive Community Treatment team whose composition ensures redundant relationship coverage. Team composition requires one mobile psychiatrist, one registered nurse, one licensed social worker, and one outreach case manager alongside a peer support specialist, a substance use counselor, and a team coordinator. To maintain operational consistency despite high sector turnover, the MDI standard mandates a 20 percent staffing buffer and a team-level identity protocol where two or more staff members share relationship responsibility for every resident. This redundant coverage prevents trust cycles from resetting when individual staff members transition; consequently, the relationship with the participant remains a property of the system rather than the individual clinician. This total of seven to eight full-time equivalents manages an active caseload ratio of 80 to 120 Pipeline C clients.

The operational protocol mandates prioritizing presence over service offers. Initial engagement deploys only coffee, food, wound care, and harm reduction supplies. Engagement operates strictly without intake forms or entry conditions. Teams establish a fixed schedule returning to the identical

individuals at the identical encampment on the identical days. This predictability structurally reduces threat-perception in paranoid individuals. Teams update HMIS records following every contact documenting behavioral changes, new anchors identified, and crisis windows observed.

The peer specialist's function is irreplaceable. Pipeline C individuals with paranoid systems can often accept contact from someone who has visibly survived the same experiences. The psychiatrist and social worker follow the peer specialist's relationship. Because the team maintains a strict non-discharge policy, service refusal is documented while outreach contact continues uninterrupted.

Engagement timeline realities indicate Pipeline B requires 4 to 16 weeks of contact before productively receiving a warm offer. Pipeline C requires 6 to 24 months of continuous engagement. Some individuals necessitate multi-year sustained contact. The ACT literature demonstrates persistence without pressure as the only evidence-based approach for this cohort (Substance Abuse and Mental Health Services Administration, 2024).

3.3. Stage 3: The Warm Offer Window

Voluntary movement from the street to housing most frequently occurs within a tight window following an acute crisis event at the encampment site including the death of a nearby individual, a violent incident, a severe weather event, or an acute medical emergency. The window closes rapidly. ACT teams maintaining sustained presence capture this window while one-time outreach teams predictably miss it.

The offer structure is specific, immediate, and irreversible. Unit assignment is pre-confirmed in HMIS before the conversation begins. The pet has a kennel unit at the veterinary node. The cart is photographed, tagged, and GPS-tracked secure possession storage accessible by Radio Frequency Identification (RFID). The social worker has prepared all intake documentation before the offer conversation including identification, benefits enrollment stubs, and the housing agreement. Because bureaucratic friction is eliminated prior to the offer, physical movement into the unit occurs within hours of acceptance.

The specificity of this offer is structural. It eliminates every rational objection Pipeline B uses and reduces the cognitive load on Pipeline C individuals who cannot process abstract future states.

3.4. Stage 4: Encampment Resolution

Encampment clearance without simultaneous housing placement produces geographic redistribution only. Municipal sweeps ignoring the by-name registry and warm offer protocol sever ACT relationships and extend engagement timelines by months.

For urban encampments, clearance operations trigger only after every individual has received a specific and immediate unit offer or enters active CARE Court or AOT supervision. This conditionality governs urban populations; riparian abatement under the Clean Water Act proceeds under independent environmental authority, though MDI provides simultaneous placement options to ensure clearance durability. Clearance functions as a logistics event where sanitation and storage teams accompany the housing placement. Post-clearance operations require ACT team follow-up with placed individuals for 90 days to prevent re-encampment.

3.5. ACT Workforce Sizing

The 2025 Los Angeles Homeless Services Authority (LAHSA) Greater Los Angeles County PIT Count recorded approximately 75,312 total homeless with fifty-eight percent unsheltered, yielding approximately 47,100 unsheltered individuals on any night when adjusted for the previously hidden riparian cohort (Los Angeles Homeless Services Authority, 2025). PIT methodology systematically undercounts the most deeply hidden encampments.

Pipeline	Percent	Count	Intervention
A. Voluntary	~26 percent	~12,246	CES rapid placement
B. Engageable	~35 percent	~16,485	Warm offer at crisis window
C. Calcified (Urban + Riparian)	~34 percent	~16,014	ACT sustained presence + legal
D. Voluntary Nomadic	~5 percent	~2,355	Harm reduction without placement

Table 3. Pipeline Distribution Applied to Unsheltered Count.

One California Plaza operating 2,000 units distributes its inventory across its target zone: approximately 800 units reserved for Pipeline C, 1,200 units for Pipeline B, and zero units for Pipeline A. The prototype does not resolve the entire 16,500-person county-wide Pipeline C demographic encompassing both urban and riparian cohorts; it requires five ACT teams targeting those specific 500 Pipeline C individuals within its geographic boundary. To manage the 12 to 24 month engagement timeline without exhausting the static 500-client capacity, ACT teams operate on a graduated step-down throughput model, rotating stabilized individuals to lower-acuity case management to admit new entrants. This operation requires 40 full-time equivalents (FTE) alongside eight to ten field registry teams requiring 25 equivalents. The ground floor clinical node staff requires six equivalents alongside 13 Pod Stewards, ten robotic swarm supervisors, and five digital access center staff. The total pilot field and residential workforce demands approximately 99 full-time equivalents.

The 20-tower, 40,000-person system demands 165 ACT teams for 16,500 Pipeline C individuals at 100 caseloads equaling 1,320 equivalents alongside 300 field registry workers, 260 Pod Stewards, 120 ground floor clinical staff, 200 robotic swarm supervisors, and 100 digital access center staff. The total full system requires approximately 2,300 full-time equivalents.

3.6. Workforce Funding Stack

The MDI field workforce draws from four interlocking revenue streams with Medi-Cal reimbursement as the load-bearing base.

ACT performs as a Medi-Cal-billable service in California under the Community Mental Health Services framework (California Department of Health Care Services, 2024a). Approximately 85 to 90 percent of the Pipeline C population qualifies for Medi-Cal (UCSF Benioff Homelessness and Housing Initiative, 2023). At the standard ACT billing rate of 150 dollars to 190 dollars per day per enrolled client alongside a single team carrying 100 clients generates a theoretical gross potential of approximately 5.5 million dollars to nearly 7 million dollars annually. Across 165 teams the structure generates a theoretical gross of 903 million dollars to 1 billion 144 million dollars annually; applying a strict 30 to 45 day annual billable window per client based on historical DHCS benchmarks to account for billing constraints (California Department of Health Care Services, 2024b) yields a functional offset of approximately 95 million dollars to 105 million dollars annually.

The Mental Health Services Act generates approximately 3.5 billion dollars statewide annually. Los Angeles County's MHSA allocation reaches approximately 400 million dollars per year (California Mental Health Services Oversight and Accountability Commission, 2024). ACT teams for the chronically homeless unsheltered rank among the highest-priority eligible expenditures under the Full Service Partnership category.

The Substance Abuse and Mental Health Services Administration (SAMHSA) funds ACT implementation through competitive grants of 500,000 dollars to 2 million dollars per program (Substance Abuse and Mental Health Services Administration, 2024). Measure Alpha covers residential infrastructure and a portion of field architecture operating costs. CARE Court implementation via AB 1976 appropriating 14.7 million dollars statewide qualifies the Stewardship Authority's petitioning function as an eligible implementation cost (California State Legislature, 2024a).

Table 4 consolidates the full-system workforce cost structure and the Medi-Cal billing offset that reduces net public expenditure.

Cost Item	Annual Gross	Medi-Cal Offset	Net Public Cost
Full ACT workforce (165 teams)	~\$99M	~\$95–99M	~\$0M
Registry/documentation (300 FTE)	~\$18M	—	~\$18M
Pod Stewards + full residential workforce (680 FTE)	~\$68M	~\$2–7M	~\$61–66M
Total field + residential	~\$185M	~\$97–106M	~\$79–84M

Table 4. Workforce Funding Stack. Net Public Cost After Medi-Cal Billing Offset. The 68 million dollar line item covers the entire 680-person full network residential facility staffing model at an estimated 100,000 dollar per-FTE burdened rate, establishing the distinct wage-to-FTE multiplier separate from field ACT capitalization.

The net public cost of 79 million dollars to 89 million dollars annually for the full 40,000-person system falls within the combined Measure Alpha and MHSA allocation available to LA County (Los Angeles County Chief Executive Office, 2025). The workforce is a deficit-neutral employment pipeline once Medi-Cal billing is accounted for. This employment pool is a labor market event for Los Angeles with a structural preference for hiring from the populations the system serves.

4. The Legal Lever System: Parallel Compelled Pathway

For Pipeline C individuals whose anosognosia or grave disability makes voluntary engagement clinically impossible, a parallel legal track operates alongside ACT outreach. This is a court-supervised clinical pathway that ensures no individual falls permanently through the gap between voluntary engagement capacity and the need for stabilization. California’s current legal architecture provides three instruments of increasing restriction.

4.1. CARE Court (The CARE Act, 2023)

All 58 California counties achieved operational status as of 2024 (California State Legislature, 2022). The target population focuses on adults with schizophrenia spectrum or other psychotic disorders alongside expanded inclusion for bipolar disorder with psychotic features as of early 2026 (California State Legislature, 2024b). Restriction levels remain low. The pathway functions non-custodially. The individual remains in the community.

Families, roommates, first responders, or healthcare providers file the petition. The Court reviews the petition and upon meeting criteria establishes a 12 to 24 month CARE Agreement operating as a court-monitored community treatment plan including housing, medication, and wraparound services. A multidisciplinary team reports to the court on progress. Persistent non-compliance triggers escalating review. CARE Court structurally diverts individuals away from more restrictive LPS conservatorships.

The MDI Stewardship Authority does not file CARE Court petitions and has no active role as petitioner in any involuntary or compelled proceeding. The Authority’s by-name HMIS registry contains documented field observations of behavioral presentation for Pipeline C individuals. This documentation exists as a factual record that authorized petitioners — families, first responders, or healthcare providers — may draw upon through their own independent processes. MDI’s role in any compelled pathway is limited to providing housing as one available option. The court, the treatment team, or the conservator selects it independently.

4.2. AOT: Assisted Outpatient Treatment

The instrument targets individuals with a documented history of treatment non-compliance and at least one of repeated psychiatric hospitalizations, incarcerations, or documented threats of or acts of violence (California State Legislature, 2002). History of system interaction acts as the primary criterion functioning distinctly from CARE Court’s diagnosis-first focus. Restriction levels hit moderate. The mechanism executes court-ordered outpatient treatment via civil court order. It carries no conviction. It mandates medication compliance and outpatient clinical engagement. Non-compliance triggers court review and potential escalation to LPS.

ACT teams accumulate documented interaction histories in HMIS. For individuals with observable histories of hospitalization, incarceration, or threatening behavior, AOT petitions may be filed by authorized parties. The Stewardship Authority does not file or co-file AOT petitions. Its HMIS documentation exists as a factual field record accessible through standard legal discovery or referral processes, not as an instrument the Authority deploys to initiate proceedings.

4.3. LPS Conservatorship (SB 43 Expansion)

The target population defines individuals deemed gravely disabled and unable to provide for their own food, clothing, or shelter due to a mental health disorder or severe substance use disorder (California State Legislature, 2023). Restriction levels rank high. A conservator is legally appointed to make medical and housing decisions on behalf of the individual. The system operates as a last resort.

Initiated following a 72-hour involuntary 5150 psychiatric hold or CARE Court escalation. Court appoints Public Guardian or designated conservator. Conservator may consent to housing placement and medical treatment on the individual’s behalf. Subject to periodic court review.

For the most profoundly calcified Pipeline C individuals, those for whom years of ACT engagement have produced no voluntary movement, LPS Conservatorship provides the legal mechanism for housing placement. A court-appointed conservator may independently select an available ALMU unit as the housing placement through normal court channels. MDI does not petition for conservatorship, does not participate in those proceedings, and does not receive a unit assignment on any individual’s behalf. The conservator acts through the court; MDI provides the housing inventory from which the court may draw.

Conservatorship is not indefinite institutionalization. The MDI ALMU environment is designed to provide the stability that allows psychiatric symptoms to stabilize over time. As condition improves,

conservatorship review may result in restoration of legal autonomy, at which point the Stewardship Contract governs tenancy voluntarily.

4.4. The Three-Instrument Decision Matrix

Table 5 maps each clinical condition to the appropriate legal instrument and the Stewardship Authority’s operational role within that pathway.

Condition	Instrument	MDI Role
Schizophrenia spectrum, no prior system history	CARE Court	Housing inventory available; no MDI petitioning role
History of hospitalization/incarceration/threats	AOT (Laura’s Law)	HMIS record as factual field document only; no petition role
Gravely disabled, cannot self-provide	LPS Conservatorship	ALMU available for independent court/conservator selection
Voluntary but treatment-resistant	ACT sustained presence	Field Architecture (no legal instrument)

Table 5. Legal Lever Decision Matrix and MDI Role.

The MDI Stewardship Authority has no active petitioning, filing, or co-petitioning role in any CARE Court, AOT, or LPS proceeding. MDI’s position relative to the legal lever system is that of a housing provider whose inventory is available for court-directed or conservator-directed placement through independent legal channels. The by-name HMIS registry remains a field documentation resource; it is not deployed by MDI as an evidentiary instrument in compelled proceedings. ACT teams continue sustained presence engagement for voluntary-track individuals throughout the duration of any parallel legal process, maintaining the voluntary pathway as the primary and preferred route regardless of whether a legal track is simultaneously active for that individual.

5. Phase Two: Ground Floor Intake and Operational Continuity

The ground floor of the MDI tower carries three spatially distinct and operationally independent systems on a single 26,000 square foot floor plate. They co-exist without mixing. The architecture enforces separation. It does not rely on staff enforcement.

5.1. The Open Resource Center

The public-facing, welcoming commons. Low-barrier, zero conditions, open to registered guests and upper-floor residents. Operations are strictly environmental with access requiring only an RFID token or thermal scan. Clinical triage and assessment are geometrically excluded from this space. New street arrivals from Pipeline A, Pipeline B, and voluntary nomadic individuals from Pipeline D access hygiene, laundry, food, or device charging voluntarily without intake preconditions. Upper-floor MDI residents use the commons as their residential amenity floor providing cafeteria, veterinary node, dog run access, Digital Access Center, and decompression zones.

Atmosphere requires circadian lighting providing warm tones alongside biophilic materials and curved surfaces eliminating institutional fixtures. The cafeteria capacity hits 400. The cafeteria operates a robotic kitchen and direct-access pharmacy vending. A real-time unit availability display shows unit availability across the tower network. The design features a veterinary node adjacent to dog run access.

5.2. Zone A: The Sally Port

Secured, clinical, dedicated arrival node for high-acuity individuals via mobile crisis team or law enforcement handoff. The node requires a separate entrance from the Resource Center and operates optically isolated from the general commons.

Interlocking double-door system prevents spillover volatility from the arrival zone into the general ground floor. Anti-ligature fixtures, tamper-resistant materials, reinforced glazing integrated into a residential-scale material palette (American College of Emergency Physicians, 2020). Clear sightlines from staff node to all positions. No blind corners.

Staffing requires a Dual-Role Crisis Node. One Crisis Intervention Team (CIT)-trained behavioral health clinician and one peer support specialist with lived experience (CIT International, 2023). The clinician manages clinical assessment. The peer specialist manages the human connection that reduces threat-perception.

The de-escalation environment deploys tunable warm spectrum circadian lighting, acoustic dampening panels, biophilic geometry with curved surfaces and no hard ninety-degree corners, and nature-sound atmospheric integration.

5.3. Zone B: Triage and Acuity Segmentation

Immediate conversational assessment to route guests into appropriate stabilization pathways. Non-clinical, trauma-informed screening. Staff observe orientation level, agitation indicators, and somatic presentation including wound status, dehydration signs, and metabolic distress. Guests partition into low-acuity flow for self-directed access to hygiene, laundry, cafeteria, and decompression zones or high-acuity support for escorted access to supervised single-occupancy hygiene nodes, wound care, and direct clinical handoff.

The real-time unit availability display shows across the tower network. It neutralizes scarcity-driven anxiety before it becomes behavioral agitation.

5.4. The Incident Pathway and Outbreak Design

Decompensation events within the Resource Center require rapid clinical intervention isolated from the general population. The ground floor does not treat outbreaks as exceptional events. The design anticipates multiple decompensation incidents per day as the baseline operational condition. Weekend and holiday periods generate elevated incident frequency correlated with reduced access to community mental health services and increased substance use in the surrounding street environment. Dedicated surge clinical staff supplement baseline ground floor personnel during these windows to manage MS-related destabilization incidents and relieve community hospitals of presentations that belong in the tower's clinical infrastructure.

A dedicated staff-key-only corridor connects the Resource Center interior directly to Zone A through a rear-wall passage. This transit route is optically blind to the general commons floor and unmarked. When staff identify an acute decompensation event, the individual is moved through the back-door corridor into Zone A without traversing the Resource Center population. The first human the decompensating individual encounters on transition is a peer support specialist with lived experience. If physical stabilization is required before transit, two designated de-escalation alcoves, visually separated from the main commons by biophilic partition walls, allow private attendance by two staff members until stability permits back-door transit. The pathway from resource center incident to Zone A crisis management does not pass through the open commons at any point.

5.5. Acuity-Segmented Hygiene

Communal hygiene spaces are primary origins of multi-actor escalation. The MDI specification eliminates communal hygiene entirely, replacing shared facilities with single-occupancy, medical-grade modules. These modules feature resident-controlled water temperature defaulting to 38 degrees Celsius and a low-lighting option with warm LEDs to reduce agitation. To prevent escalation driven by urgency, residents face no time pressure for the first 30 minutes while UV-C and HEPA air scrubbing maintain air safety.

Clothing exchange for Diogenes-pattern individuals functions as a supplemental addition. Hygiene kit is offered as a gift. Staff bag and tag soiled clothing pending consent for disposal. Skin and wound assessments are conducted immediately post-hygiene. Chronic wound pain is identified as a primary driver of agitation and pain management precedes psychiatric medication adjustment in the stabilization sequence.

5.6. Intake Stabilization Sequence

Hours 0 through 4 prioritize rehydration. Oral rehydration requires World Health Organization formulated electrolyte solutions (World Health Organization, 2006). Rapid water loading in severely dehydrated individuals with depleted sodium produces hyponatremia, a potentially fatal cerebral edema. This is a life-safety specification that drives robotic kitchen and pharmacy protocols.

Hours 4 through 24 focus on nutritional triage. Refeeding Syndrome is a life-threatening risk for severely malnourished arrivals (Mehanna et al., 2008). Rapid reintroduction of carbohydrates triggers electrolyte shifts producing cardiac arrhythmia and neurological complications. The cafeteria requires a clinical triage layer identifying high-risk individuals indicating Body Mass Index (BMI) ≤ 16 , visible severe wasting, or reported days without food and routing them to a gradual reintroduction protocol before open cafeteria access.

Hours 24 through 72 comprise the psychiatric assessment window executing at the scattered-site MHRC cluster following secure transport from the resource center. Once metabolic stability is achieved within the resource center, the individual's neurological baseline rises enough for meaningful psychiatric evaluation. Attempting psychiatric assessment before this window produces false-negative capacity and compliance assessments. Psychiatric assessment follows metabolic stabilization.

For guests requiring immediate psychiatric stabilization beyond resource center capacity, the architecture provides direct-path access to an on-site psychiatric emergency node.

6. Phase Three: Ontological Architecture

Ontological security is the provision of structural predictability sufficient to eliminate the routine of being rousted (Giddens, 1991; Laing, 1960). Its core markers mandate a space for daily routines alongside a sense of privacy, control over the environment, and the ability to maintain a coherent sense of self away from external intrusion. The ALMU achieves engineering designed to deliver all four markers simultaneously.

6.1. The Acoustic Sanctuary: STC 65

Sound Transmission Class (STC) 55 is the biological floor. It prevents physiological hearing damage. STC 65 is the ontological threshold. It neutralizes the vibration of the street and neighboring micro-movements that trigger hyper-vigilance in trauma-impacted residents (Frontiers in Psychology Editorial Board, 2024; GB Acoustics, 2024).

Construction demands decoupled mass assembly securing resilient channels isolating gypsum layers from structural framing. The build specifies double-layer 5/8-inch Type X gypsum featuring Green Glue viscoelastic damping compound between layers. Teams apply acoustic caulk at all penetrations. Floor-ceiling assemblies incorporate a floating floor system on rubber isolators over a concrete substrate. STC 60 and above renders loud speech inaudible in adjacent spaces. STC 65 reduces ambient sound transfer to levels approaching clinical quiet-room standards.

Active masking layer integrated into the 6-to-10 exchange per hour air filtration system. Biological white-noise frequencies including running water and wind spectra introduced at low amplitude. Neutralizes residual micro-sound without the clinical sterility of silence, which can itself be disorienting for individuals habituated to outdoor ambient environments.

6.2. The ALMU: Sovereign Cabin

A precision-engineered 150 square foot module provides a locked door controlled exclusively by the resident.

The wet bath module occupies 24 to 28 square feet. Wet room design is a single waterproofed chamber where the shower pan is the entire bathroom floor. Toilet, wall-mounted sink, and showerhead share a drain-sloped floor. Resident-controlled water temperature. Solar thermal pre-heated supply. The architecture prohibits communal hygiene facilities. Every ALMU incorporates an independent wet bath. This is a non-negotiable dignity threshold.

The living volume occupies 90 to 92 square feet. Murphy wall-bed folds flat when stowed, reclaiming approximately 35 square feet of floor during daytime. Wall-mounted fold-down desk with seated and standing dual configuration. Under-bed and under-desk locking storage. Countertop one-point-seven cubic foot mini-refrigerator, induction cooktop plate, microwave alcove. No gas range. Fire safety constraint at this density is absolute. Built-in wardrobe column measuring 12 inches deep by 24 inches wide.

The remaining 20 to 30 square feet accommodate the demising wall thickness, acoustic insulation layers, and the integrated mechanical, electrical, and plumbing chases required for the STC 65 specification and individual HVAC control. The 150-square-foot ALMU represents the net private dimension. The gross residential footprint per unit consumes 400 square feet, allocating the 250-square-foot delta to primary circulation corridors, the Tier 1 pod-level biophilic nodes, the pod common kitchens, elevator lobbies, and centralized mechanical chases. Across 2,002 units, this gross distribution mathematically accounts for 800,800 square feet. The remaining 199,200 square feet accommodates the four-story commercial base and the ground floor triage architecture consuming 104,000 square feet, explicitly allocating the remaining 95,200 square feet across the five Tier 2 tower-level sanctuaries at 19,040 square feet per level.

Materials feature a warm-toned material palette. Matte birch ply paneling meets stone-texture wall surfaces. The spec eliminates white institutional finishes. Systems feature individual Heating, Ventilation, and Air Conditioning (HVAC) control within a 68-to-78-degree operating band. Facilities execute clinical-grade HEPA and UV-C scrubbing. Air quality exceeds average Los Angeles apartment standards.

The hard door demands solid-core construction integrating an STC 65 acoustic frame and threshold seal. Installations include a quiet-operation digital lock. The ground floor Zone B triage node executes immediate keycard issuance. Sovereignty of the private space is guaranteed from the moment of unit assignment.

The window priority rule dictates all ALMU units receive perimeter-adjacent positions first. Interior floor space converts to pod common areas, biophilic nodes, and pod kitchens. Where interior units prove necessary to meet the 2,000-unit target, full-spectrum circadian LED ceiling panels execute a programmable dawn-to-midday-to-dusk 24-hour cycle providing documented mitigation of seasonal affective disorder and circadian disruption. Perimeter units provide full-height commercial

glazing with floor-to-ceiling sky views. Unobstructed sky access remains a therapeutic variable for populations exiting a street encampment.

Module definition as a transitional medical facility establishes a legal structure designed to bypass housing court jurisdiction while preserving residential dignity. By strictly capping continuous occupancy below 30 days, residents cannot claim statutory tenancy under California Civil Code Section 1940. The Stewardship Contract governs all short-term tenure conditions.

Redundant resilience requires hydrogen or dual-fuel backup generators. Architecture demands a 96-hour potable water cistern mandate holding 560,000 gallons total at One California Plaza distributed across 35 percent of the P3, P4, and P5 basement slabs to maintain standard 8-to-10-foot vertical parking clearances. Pressurized sewage ejectors carry a 50,000-hour Mean Time Between Failures (MTBF) mandate.

6.3. Dunbar Verticality: Partitioning Against Anonymity

A 2,000-unit tower is institutionally scaled, too large for organic social cohesion. Dunbar's Number establishing a limit of approximately 150 represents the cognitive ceiling for stable, trust-based social relationships (Dunbar, 1992). Above 150, organic mutual accountability fails, necessitating formal enforcement mechanisms.

The tower mass partitions into 13 independent Home Pods of 154 units each (yielding 2,002 units, with two units permanently reserved as an administrative vacancy buffer to sustain a rigid 2,000-unit operating baseline). While this structural allocation marginally exceeds the precise Dunbar Number limit of 150, it falls well within the natural empirical variance of Dunbar's research. These pods function as operationally autonomous zones with dedicated staff clusters, localized incident containment protocols, and independent vertical circulation loops serviced by the 28-elevator bank. By programming independent elevator loops, Pod 3 residents encounter primarily their own pod neighbors in transit, thereby creating recognizable daily familiarity. Digital lockout enforces pod boundaries on shared amenity floors via RFID-linked time-slot scheduling, which prevents anonymous mass congregation and ensures that any behavioral or infrastructure failure remains contained within a single pod. This modular architecture transforms the 2,000-unit tower from a single point of failure into a network of 13 semi-autonomous residential communities.

Dunbar Group sizing within each pod is not standardized. The ITP finalized at day 30 of Phase Zero specifies individual DG assignment based on observed social tolerance, agitation triggers, and psychiatric presentation. Some residents stabilize best in smaller DG clusters of five to eight individ-

uals. Others integrate readily into groups of twenty to thirty. The Dunbar number of 150 establishes the pod-level ceiling, not the individual's social density threshold. Smaller DG assignment is not a lower-status outcome. It is a clinically appropriate match to the individual's neurological needs at the point of ALMU placement. DG assignment is revisable as the resident stabilizes and social tolerance expands.

6.4. Biophilic Infrastructure: Two-Tier System

Biophilic exposure reduces cortisol, improves mood regulation, and decreases agitation incidents in high-density residential settings (Grand Rising Behavioral Health, 2024; IMEG Corp, 2024). The MDI tower delivers biophilic access at two distinct scales.

Tier 1 Pod-Level Node occupying every pod and every 4 to 5 floors. A 200 to 300 square foot planted alcove embedded in the path residents travel daily. Living plant wall with automated drip irrigation. Window-adjacent seating. Small conversation-scaled table. The living plant wall absorbs mid-frequency sound, creating a quieter micro-zone within the circulation path. Native California species, drought-tolerant, low-volatile organic compound (VOC), sensory-appropriate. Positioned immediately adjacent to the pod common kitchen. Cooking smells, plant presence, natural light, and informal social encounter occupy the same zone. Residents traverse the pod node organically. The five to ten minute passive biophilic exposure that occurs incidentally during daily movement delivers the documented cortisol-reduction dose without requiring behavioral change.

Tier 2 Tower-Level Sanctuary occupying every seven floors. A destination. Oxygenated lounge with native California flora, operating at 50 percent of original HVAC design load surplus. Double-height volume or exterior terrace where structural conditions permit. Hydroponic garden nodes with resident-accessible cultivation plots. Rooftop greenspace with automated irrigation and weather-protected seating. This is where residents from different pods encounter each other. Cross-pod social bridges form here organically, reducing the risk of pod insularity becoming pod tribalism.

7. Social Infrastructure: The Human Layer

The tower's physical architecture creates the conditions for ontological security. The social infrastructure sustains it. Architectural design engineers the spatial conditions that precipitate community formation (Frontiers in Psychology Editorial Board, 2024).

7.1. The Pod Steward: The Gardien Model

The French Gardien model in social housing demonstrates that a live-in, on-site human presence functions as a Social Broker, bridging anonymous institutional management and the daily human realities of residents (Harvard Graduate School of Design, 2023; Eindhoven University of Technology, 2024).

Security guards execute enforcement protocols which residents perceive as a threat signal, particularly Pipeline C individuals with paranoid presentations. A Pod Steward executes recognition protocols. They know every resident by name, observe behavioral baseline, and respond to deviation exclusively as a welfare signal.

Staffing requires a credentialed peer support specialist possessing certificated lived experience of homelessness or psychiatric illness operating with social work supervision. One Pod Steward serves each Dunbar Pod of 154 individuals. Residential unit within the pod preferred. The 24-hour welfare check protocol requires zero movement detection for 24 hours on the life-safety telemetry prompting a Pod Steward face-to-face check-in. The human layer interprets the data signal and responds with judgment and relationship.

7.2. Resident Civic Engagement Pathway

Voluntary participation tracks that transition residents from passive service recipients to active stakeholders. Participation is voluntary. Participation earns Stewardship Status, which carries social recognition within the pod.

Garden Stewardship requires care of the pod-level biophilic node and tower-level sanctuary. Horticultural therapy research documents measurable psychiatric stabilization outcomes from plant care responsibility. Robotic Swarm Supervision mandates residents shadow and assist the Gausium platform maintenance cycle reducing paranoid threat-perception of automated systems and granting agency over the physical environment. Community Kitchen Coordination involves pod-level meal scheduling alongside shared cooking and informal hosting. Food preparation functions as social

ritual. Digital Access Center Support invites residents with vocational or technical skills to mentor others in peer-to-peer skill transfer.

7.3. The Pod Common Kitchen

The ground floor cafeteria handles mass nutrition operating a four-hundred-person capacity robotic kitchen. The pod common kitchen handles the informal, voluntary, social act of cooking together. Whereas the cafeteria is engineered to provide mass metabolic sustenance, the pod kitchen functions strictly to build the social architecture of the pod through shared meal preparation.

Scale accommodates 8 to 15 people cooking and eating together. Four-to-six-burner induction range. Two ovens. Commercial-grade refrigerators. Prep island with seating for six. Dining table for 10 to 12. Open shelving for communal pantry, restocked from the Resource Bank inventory via Pod Council coordination.

Placement must sit immediately adjacent to the pod-level biophilic node. The kitchen and the planted alcove share a visual and olfactory field. This adjacency is intentional. It is the warmest point in the residential pod. No booking required for informal use. Always unlocked during waking hours spanning 0600 to 2300. The Nocturnal Access Protocol does not apply to the pod kitchen.

7.4. Animal Companions: The On-Site Facility Model

Pipeline B's primary stated reason for refusing shelter is pet separation. The MDI architecture guarantees contiguous pet ownership and unlimited daytime access.

154 residents in a Dunbar Pod with each person's animal companion present in their unit would make the environment intolerable for the majority. Dogs bark, shed, and have elimination accidents. Dander saturates shared HVAC corridors. Inter-animal aggression in elevators is a safety event at this density. The residential floors are for human habitation. Animal companions live in the on-site facility.

Professionally managed kennel suites with individual enclosed runs. Climate-controlled, sound-dampened, odor-managed. Resident access remains unlimited during facility hours spanning 0600 to 2200. Zero cost to residents. On-site non-profit veterinary partnership provides all medical, vaccination, behavioral, and spay-neuter services.

Small caged animals (birds, hamsters, fish) may be registered for in-unit accommodation. No in-unit cats or dogs on residential floors. This is a fixed rule.

7.5. The Residential Behavioral Floor

Alcohol is permitted in the individual ALMU. Entry operates independent of sobriety. This is Housing First doctrine (Y-Foundation, 2023). The behavioral floor activates at the point of externalized impact. Intoxication becoming a safety event in shared corridors triggers the de-escalation protocol directing Pod Steward response and peer specialist engagement. Whereas consuming alcohol within the private unit remains a protected liberty, generating a safety event in shared communal space immediately triggers the de-escalation protocol.

The internal Stewardship Contract prohibits indoor smoking within the facility to eliminate fire risk and protect the STC 65 ventilation standard. The design response deploys a purpose-designed, covered, weatherproofed outdoor smoking terrace with seating, weather protection, and social adjacency to the ground floor commons. Nicotine replacement therapy remains available at the pharmacy node. Voluntary.

The ALMU residential track is not the immediate destination for individuals in active, unmanaged heavy substance dependence that is functionally incapacitating. These individuals require a Managed Alcohol Program facility or medically supervised transitional care setting first (Podymow et al., 2006; Larimer et al., 2009). Post-medical stabilization, transfer to the ALMU track follows. The legal lever system can mandate MAP-equivalent treatment as a clinical condition. The ALMU is the destination after stabilization.

7.6. The Three Ps Infrastructure

The three primary barriers to shelter acceptance and their MDI architectural responses are specified in Table 6.

Barrier	MDI Response	Location
Pets	On-site facility granting unlimited daytime access alongside veterinary node and dog run	Level 2 facility
Partners	Co-location within same Dunbar Pod; Two adjacent standard 150 sq ft ALMUs	Intake matching, HMIS
Possessions	Photographed, tagged, GPS-tracked secure storage, 24/7 RFID access	Ground floor storage

Table 6. Three Ps Infrastructure: Structural Elimination of Shelter Refusal Barriers.

Eliminating all three barriers simultaneously converts the MDI offer from a generic shelter referral into a structurally rational proposition that the individual can accept without sacrificing survival anchors.

7.7. Digital Access Center

Low-distraction, high-bandwidth vocational and educational access. Gigabit-class internet. Hardware lending includes laptops, tablets, and peripherals. Vocational training via non-profit partnerships. Legal self-defense resources feature benefits enrollment, eviction defense, ID restoration, credit rehabilitation. Gig-work platform access enabling income generation while in residence. Non-profit Technical Assistance Hub operates as permanent administrative partner in the node space.

7.8. Micro-Community Governance

The Pod Council convenes a voluntary monthly pod-level meeting where residents collectively identify infrastructure concerns alongside common space scheduling and social friction for mediation. The Pod Steward facilitates. Residents chair. The Digital Pod Channel provides privacy-first opt-in communication within the pod. Pod Mapping function allows residents to voluntarily signal capacity to assist neighbors or request support. No Authority visibility into channel content.

The Tenancy Bridge Guarantee ensures the Stewardship Contract remains active as a legal guarantor of tenure until a new private market lease is verified, signed, and funded. No resident transitions to the independent market without a confirmed landing. The contract is the safety net beneath the transition.

8. The Cybernetic Grid: Automated-Human Operational Balance

The MDI cybernetic framework operates on a single governing principle requiring automation for the machine and agency for the human. Total automation is a form of institutional neglect. It replaces human judgment with algorithmic pattern-matching in the situations where pattern-matching fails most catastrophically. Total human management at 2,000-unit scale is operationally impossible. The MDI operational balance delegates technology to the mechanical layer while positioning humans as the relational layer.

8.1. The Telemetry Threshold

Smart home sensor deployment in low-income housing produces a Panopticon Effect. Residents alter behavior under perceived observation, experience performance stress within their own homes, and exhibit progressive erosion of the home as a psychological sanctuary (NYU Marron Institute of Urban Management, 2024; University of Southern California, 2024). This undermines ontological security.

The residential data minimization protocol mandates the only data exported from a residential ALMU unit to the central grid is a binary safety indicator establishing Safe or Not Safe. All other environmental data, such as sound levels, movement patterns, and temperature, is processed locally and purged within 24 hours. The data architecture structurally precludes behavioral profiling.

Authorized telemetry forming the Metabolic Substrate includes smoke and CO2 detection, water leak and pipe pressure monitoring, air quality checks measuring PM2.5 and CO2 concentration for HVAC optimization, utility flow at building system level, structural vibration for anti-stripping detection, sewage ejector pressure and MTBF tracking, and the life-safety binary occupancy check where zero movement for 24 hours triggers Pod Steward notification.

Telemetry permanently unauthorized covering the Social Substrate includes internal movement within the ALMU, sound levels inside the ALMU, guest visitors or behavioral patterns, consumption patterns at unit level, and digital activity on resident devices.

8.2. CPTED: Natural Surveillance Over Digital Surveillance

Crime Prevention Through Environmental Design is the architectural alternative to camera-dense digital surveillance (Council on Tall Buildings and Urban Habitat, 2024). Its core principle demands designing spaces enabling residents to naturally observe common areas as part of daily movement.

Glass-partitioned common rooms along primary circulation loops. High-visibility stairwells with clear sightlines to landings. The Digital Access Center and pod kitchen positioned at circulation chokepoints, ensuring continuous human presence without surveillance. Landscaping and pod garden placement follows the 3-foot/10-foot rule limiting visual obstruction below three feet alongside establishing clear sightlines above ten feet.

Digital camera placement stands strictly limited to building perimeter capturing exterior entries and exits alongside the Sally Port Zone A and bike storage node. No corridor cameras in residential pods. Corridor cameras elevate resident anxiety while failing to reduce incident rates.

8.3. The Robotic Platform: The Gausium Scrubber 50

The Gausium Scrubber 50 is ISO 3741 compliant with 80 percent water recycling efficiency, automated maintenance of common floor surfaces, corridor cleaning, and rapid response to biological waste events.

The Swarm Integration Protocol mandates robotic swarms perform rapid replacement of standardized components including door lock hardware, panel seals, and utility node covers exclusively under Pod Steward authorization. Systems execute proactive maintenance addressing environmental envelope integrity and leak detection through pipe repair fully autonomously. Reactive maintenance requiring response to biological events in residential areas necessitates Pod Steward human authorization for dispatch.

8.4. Nocturnal Access Protocol

Automated throttling of common floor access between 0000 and 0600 hours. Gym, cafeteria, and social nodes shift to reduced-capacity mode. Primary elevator banks prioritize emergency and direct-floor access only. Residential behavior normalization mandates reducing unrestricted common floor activity in the overnight window to mirror the social rhythm of market-rate residential buildings and reduce the perception of an institutional facility. Medical, mental health, and safety access remain unrestricted globally. The protocol applies only to discretionary amenity access.

8.5. Transportation Asset Management

Automated Storage and Retrieval System for bicycles at ground floor and per-pod access nodes. Frame integrity and weight scan on every entry and exit cycle detects component stripping attempts. Bicycle stationary in private node beyond 72 hours triggers Pod Steward welfare check. The real-time

unit availability display shows transportation asset availability across the pod network. Scarcity-driven hoarding behavior is structurally neutralized by visible abundance.

8.6. Measure Alpha Cybernetic Allocation

1.587 percent of the 843 million dollar proposed FY 2026–27 Measure Alpha allocation yields 13,380,000 dollars annually for telemetry hardware, robotic platform maintenance, and cybernetic infrastructure (Los Angeles County Chief Executive Office, 2025). Telemetry data architecture reviewed annually by an independent Privacy Audit Board with resident pod council representation. No data retention beyond 24-hour local purge without explicit resident consent and Board approval.

9. Fiscal Architecture and the Efficiency Surplus

9.1. Layer 0: Medi-Cal Sub-Acute Billing Through MHRC Licensure

The fiscal architecture begins at Phase Zero, before any individual enters the tower. Each MHRC sub-16-unit cluster is licensed under DHCS Form 1813 and bills Medi-Cal as a specialty mental health sub-acute facility at approximately 50 percent federal financial participation through FMAP (California Department of Health Care Services, 2024c,a). A 60-day Metabolic Stabilization episode billed at the sub-acute specialty mental health services rate generates approximately 15,000 dollars per enrolled resident, of which approximately 7,500 dollars derives from federal FMAP. This layer funds the most clinically intensive phase of the pipeline. It operates under existing California law. No legislative action and no speculative waiver is required for activation. The architecture targets the Institution for Mental Disease Exclusion by isolating operations into 15-unit clusters. Because the Centers for Medicare and Medicaid Services interprets coordinated clusters operating within the contiguous 1-mile footprint of the defined geographic zone as a de facto single institution (Centers for Medicare & Medicaid Services, 2017), the fiscal model requires a preemptive Section 1115 demonstration waiver to maintain Medi-Cal eligibility.

The cost comparison is direct. A high-frequency emergency room user cycles at 3,000 to 5,000 dollars per visit across 8 to 12 visits annually, generating 24,000 to 60,000 dollars in healthcare expenditure per year with no stabilization outcome. A 60-day MHRC episode at 15,000 dollars total produces clinical stabilization enabling every downstream pipeline stage. The Medi-Cal case to county fiscal officers and DHCS reviewers is built on this comparison.

9.2. The Four-Layer Funding Stack

The MDI fiscal model operates through four reinforcing layers. Layer 0 funds Phase Zero Metabolic Stabilization through MHRC Medi-Cal sub-acute billing. Layer 1 funds ongoing tower-resident clinical services through Medi-Cal ACT billing at 150 to 190 dollars per enrolled client per day (California Department of Health Care Services, 2024a). Layer 2 funds field architecture, Pod Stewards, and residential management through Measure Alpha and MHSA Full Service Partnership allocations. Layer 3 generates the Efficiency Surplus through municipal service cost elimination, recovering the capital investment within the verified timeline. The Section 1115 Demonstration Waiver secures Layer 0 yield. Without it, the clustered Phase Zero facility design triggers the Institution for Mental Disease Exclusion. Layer 1 yield remains secure because the permanent supportive housing classification bypasses the institutional threshold entirely.

The MDI fiscal model leverages a central insight establishing the inefficiency of the current system as the operational funding source. The delta between what Los Angeles currently spends on managing chronic homelessness equaling 50,000 dollars per person per year through fragmented emergency services, law enforcement, healthcare, and judicial costs versus what the MDI system costs to operate per resident per year generates an Efficiency Surplus sufficient to recover the full capital investment within 29 months of tower activation (UCSF Benioff Homelessness and Housing Initiative, 2023; CalMatters, 2026). While the 50,000 dollar baseline utilizes a secondary public policy aggregation, it aligns with primary cost studies of high-acuity cohorts.

The financial architecture simultaneously resolves the metropolitan homelessness crisis and the downtown commercial real estate collapse.

9.3. The Commercial Real Estate (CRE) Bailout Logic

One California Plaza operates in receivership following a 300 million dollar default. Market-verified floor valuation sits at 120 million dollars equaling approximately 120 dollars per square foot (GlobeSt, 2025). This represents an approximately 80 percent discount from peak CRE valuation.

The MDI Stewardship Authority targets distressed trophy towers at the 120 dollar per square foot floor. Acquisition via Sovereign Acquisition through Receivership or by settling defaulted debt through bankruptcy bidding using state police power authority. The acquisition mechanism secures an orderly resolution of defaulted debt at a verified floor value for bondholders while converting stranded commercial inventory into stabilization infrastructure at an 80 percent discount to replacement cost (Estrada, 2026; Allwork Space, 2026).

The sixteen million square foot network is gated by binary proof of concept at One California Plaza. Successful Singular Prototype Threshold demonstration unlocks replication across the 20-tower network using the same receivership acquisition model across the distressed downtown corridor.

9.4. Prototype Capitalization

The One California Plaza prototype requires a total capital expenditure of 195 million dollars across three cost categories, detailed in Table 7.

Expense Category	Metric	Estimated Cost
Structural Acquisition	1,000,000 sq ft at \$120 per sq ft	\$120,000,000
Residential Module Installation	2,000 units at \$25,000 per unit	\$50,000,000
Physical Plant Hardening	STC 65, HVAC, Cisterns, Generators	\$25,000,000
Total Prototype Capital	Singular Financial District Tower	\$195,000,000

Table 7. *Prototype Capital Expenditure. One California Plaza. Fiscal Year 2027.*

The capital investment generates a measurable annual public efficiency return, quantified in Table 8.

Metric	Best Case	Most Probable	Worst Case
Gross Municipal Savings	\$100,000,000	\$75,000,000	\$60,000,000
Net Recurring Expenses	−\$18,000,000	−\$22,500,000	−\$27,000,000
Net Efficiency Surplus	\$82,000,000	\$52,500,000	\$33,000,000
Capital Recovery Timeline	29 Months	45 Months	71 Months

Table 8. *Efficiency Surplus Variance Matrix. Single Tower Capital Recovery.*

9.5. The Efficiency Surplus and Operational Tolerance

The municipal system spends 50,000 dollars per chronically unsheltered individual annually across emergency, psychiatric, judicial, and sanitation operations (CalMatters, 2026). Accommodating 2,000 individuals yields a best-case avoided cost of 100 million dollars annually. The most probable scenario achieves an avoided cost of 75 million dollars annually. Deducting the estimated 22.5 million dollar net recurring expense per tower yields a net mode surplus of 52.5 million dollars for the most probable execution.

The 195 million dollar single-tower capital expenditure recovers through this surplus mapped across three execution scenarios based on the 50,000-dollar avoided cost baseline. The Best Case scenario achieves full capital recovery within 29 months. The Most Probable scenario incorporates 25 percent system friction achieving recovery within 45 months. The Worst Case scenario modeling severe friction attains recovery within 71 months. Sensitivity analysis indicates that if municipal avoided costs drop to 30,000 dollars per person, the Most Probable capital recovery timeline extends to approximately 62 months.

These calculations execute before accounting for Medi-Cal ACT billing revenue, Measure Alpha allocation, or MHSA funding streams. The surplus stands self-reinforcing. Each additional tower activation multiplies the annual surplus linearly while sharing fixed administrative overhead across the network.

9.6. Measure Alpha Integration

The 843 million dollar proposed annual Measure Alpha allocation funds homelessness services across Los Angeles County (Los Angeles County Chief Executive Office, 2025; Los Angeles County Homeless Initiative, 2025). The MDI tower network operationalizes this funding by providing a verifiable infrastructure destination. Measure Alpha expenditures currently disperse across hundreds of independent providers with limited outcome verification. The MDI Stewardship Authority centralizes reporting, converting Measure Alpha into capital infrastructure investment with measurable per-resident outcomes.

9.7. Non-Profit Co-Location Economics

Floors two through four of each tower dedicate commercial-grade floor plates to non-profit partner offices providing legal aid, vocational training, behavioral health clinics, peer support organizations, substance use treatment providers, and education programs. This co-location neutralizes the geographic transportation barrier driving the majority of service defaulting in the legacy system.

Non-profit partners lease at below-market rates offset by Measure Alpha pass-through contracts. The four-to-one civilian-to-resident interaction ratio established by the Crosstown Concourse model (Crosstown Arts and Crosstown Concourse LLC, 2017) functions through non-profit staff, civilian visitors, and ground-floor public commons activation.

9.8. Seven-Stakeholder Value Hierarchy

The MDI financial architecture generates value for every constituency simultaneously. Distressed debt resolves orderly for CRE bondholders at verified floor prices. Municipal government secures a verifiable reduction in chronic homelessness causing corresponding emergency service cost elimination. State government achieves accountability-targeted deployment of the massive homelessness expenditure stream (California State Auditor, 2026). Non-profit partners acquire co-located physical presence eliminating transportation barriers. Residents receive sovereign, permanent, and dignity-preserving housing. Taxpayers see deficit-neutral operations through Medi-Cal billing offsets alongside massive

annual public efficiency returns per tower. The labor market adds substantial full-time equivalents at full system scale featuring structural hiring preference for individuals offering lived experience.

10. Legal Governance: The Stewardship Authority

The MDI Stewardship Authority operates as a state-chartered and quasi-governmental entity modeled on the Tennessee Valley Authority (United States Congress, 1933). It operates wielding delegated state police power, independent bonding capacity, and a charter mandate limited to two specific functions including acquisition and conversion of distressed commercial real estate into permanent stabilization infrastructure and operation of the MDI pipeline from field architecture through residential permanence.

10.1. The TVA Model: Operational Autonomy

The Authority is chartered by the California Legislature as a special-purpose entity with statutory independence from municipal government. This independence resolves the coordination failure that paralyzes Los Angeles homelessness policy. LAHSA, the City of Los Angeles, LA County, the Housing Authority, and scores of independent service providers operate overlapping mandates with no single entity possessing both operational authority and capital acquisition power. The Stewardship Authority consolidates field operations, legal lever petitioning, real estate acquisition, residential management, and outcome verification under a single entity with a single mandate.

Board composition requires Governor appointment with confirmation by the Senate Rules Committee establishing fixed terms preventing concurrent municipal office holding. The budget derives from Measure Alpha allocation, Medi-Cal ACT billing revenue, MHSA Full Service Partnership funds, and bond issuance executing against projected Efficiency Surplus revenue.

10.2. The Home Rule Stabilization Ordinance

Municipal zoning resistance is the primary political obstacle to MDI tower activation. The Home Rule Stabilization Ordinance pre-empts conflicting local land use regulations for qualifying Stewardship Authority projects (Holland and Knight, 2025). State housing law already supersedes conflicting local ordinances. The Ordinance codifies this supremacy into automatic by-right conversion authority for commercial structures acquired through the Sovereign Acquisition process.

SB 330 vesting locks zoning entitlements at the date of application for qualifying residential projects (Los Angeles City Planning, 2026b). Once the Stewardship Authority files a conversion application, no subsequent zoning amendment can alter the entitlement. The Adaptive Reuse Ordinance February 2026 expansion eliminates minimum unit size requirements and public hearing mandates for qualifying

conversions (Los Angeles City Planning, 2026a). The MDI ALMU at 150 square feet qualifies under the expanded ARO.

10.3. Sovereign Acquisition via Receivership

Commercial buildings in receivership following mortgage default enter a court-supervised disposition process. The Stewardship Authority participates in bankruptcy proceedings as a qualified governmental bidder with statutory acquisition priority for structures meeting MDI conversion criteria. The acquisition price targets market-verified floor valuations typically 70 to 85 percent below peak commercial real estate valuations for distressed downtown assets (GlobeSt, 2025).

The acquisition mechanism simultaneously resolves bondholder debt through orderly liquidation at floor value alongside converting stranded commercial inventory into public infrastructure supplying stabilization housing at replacement cost discounts exceeding 80 percent. This dual resolution creates political support from constituencies that normally oppose public housing expansion.

10.4. The Stewardship Contract

The Stewardship Contract governs all residential tenure within the tower. It is classified as a transitional medical occupancy. This classification mirrors hospital bed licensure precedents (California Department of Housing and Community Development, 2024). By enforcing mandatory transfers before the 30-day statutory tenancy threshold established in California Civil Code Section 1940, it is the structural innovation permitting the MDI model to operate outside standard housing court jurisdiction while preserving residential dignity protections.

The resident holds a Stewardship Contract granting transitional occupancy, privacy, and non-interference rights during the medical stabilization window. Because occupancy cannot legally exceed 29 continuous days, the resident does not hold a property interest or statutory tenancy triggering housing court eviction jurisdiction. The Stewardship Authority retains the ability to relocate a resident within the network via tower-to-tower or pod-to-pod transfers through the Rapid Transition Protocol without initiating judicial proceedings. This is the mechanism that permits management of acute behavioral crises without either punitive eviction or enduring unsafe conditions for neighboring residents.

Tenure stands transitional, capped strictly at 29 continuous days to preserve the medical facility exemption under California Civil Code Section 1940. The Stewardship Contract governs this transitional window. The only available exits include voluntary graduation to the independent market

requiring a verified Tenancy Bridge Guarantee, voluntary transfer to another MDI node to reset the occupancy clock, involuntary transfer via the Rapid Transition Protocol, or death.

LEGAL RISK ACKNOWLEDGMENT — Stewardship Contract Classification The classification of the Stewardship Contract as a “non-property interest” is a legal design choice intended to operate the MDI residential environment outside standard housing court eviction jurisdiction. This classification has not been tested in litigation and carries foreseeable legal risk. Courts in most jurisdictions examine the substantive character of a living arrangement — including the degree of exclusivity, continuity, and residential use — rather than the label a contract assigns it. A resident who occupies an ALMU unit with privacy rights, continuous tenure, and non-interference protections equivalent to a standard lease may be found by a court to hold a tenancy interest regardless of contractual nomenclature. California courts have applied this substance-over-form analysis in analogous contexts involving residential hotels, co-living arrangements, and program-conditioned occupancy agreements.

A second risk concerns statutory tenant protections. California law restricts a landlord’s ability to contract away certain statutory rights, including just-cause eviction protections under AB 1482 and applicable local rent stabilization ordinances. A court could find that these protections attach to Stewardship Contract residents regardless of how the agreement characterizes the occupancy.

This paper offers the Stewardship Contract mechanism as a legally untested design requiring further legal review before implementation. The policy architecture is sound as a design proposal. Its legal enforceability in California courts, and its interaction with state and local tenant protection statutes, must be evaluated by counsel before the Authority seeks to rely on the non-property-interest classification in any administrative or judicial proceeding. This acknowledgment applies to all fully voluntary residents and is independent of the compelled-pathway issues addressed in Section 4.

10.5. The Rapid Transition Protocol

When a resident’s behavior creates conditions of physical danger for neighboring pod members and de-escalation has failed, the Rapid Transition Protocol transfers the individual to a different pod or tower within 24 hours. Medical necessity review by the on-site psychiatric team precedes transfer. The Protocol executes a protective relocation. The resident retains their Stewardship Contract. They receive equivalent accommodations in the receiving pod. HMIS tracks the transfer and notifies the receiving Pod Steward.

This mechanism replaces eviction by ensuring that a behavioral crisis triggers immediate structural relocation to a secondary pod, thereby guaranteeing continuous housing tenure.

10.6. Federal Supremacy Leasing

For commercially leased tower space housing federal agency partners (Veterans Administration clinical outreach, Social Security Administration benefits enrollment, SAMHSA-funded programs), Federal Supremacy Leasing invokes federal preemption of conflicting municipal regulations for space occupied by federal agencies or federally funded programs. This creates a secondary legal shield for federal service co-location floors against municipal obstruction.

10.7. Section 1115 Demonstration Waiver

The Stewardship Authority petitions the California Department of Health Care Services for a Section 1115 Demonstration Waiver exclusively for Phase Zero scattered clusters to bypass the IMD threshold for short-term stabilization (Centers for Medicare & Medicaid Services, 2023). The 2,000-unit sovereign cabin tower is legally classified as independent permanent supportive housing. This classification ensures residents maintain indefinite Medicaid eligibility without triggering institutional treatment restrictions, while the external Phase Zero sites process clinical intake billing independently.

11. Comparative Models and Evidence Base

The MDI framework does not derive its architecture from theory. Every major design decision maps to a demonstrated precedent from a metropolitan, national, or clinical case study.

11.1. Houston: The Way Home Validation

Harris County deployed a coordinated metropolitan field architecture serving approximately 10,000 individuals over 13 years (Coalition for the Homeless of Houston/Harris County, 2024). The core mechanism demands a real-time by-name HMIS registry integrating over 100 partner agencies. Encampment-specific Housing Surge Events coordinate placement offers at targeted sites. The city executes clearance only after confirming housing placement. Ninety-day post-placement follow-up remains standard protocol.

Measured outcomes feature a 60 percent reduction in overall homelessness between 2011 and 2020. The 2018 Wheeler Encampment clearance placed 58 percent in permanent supportive housing. The 2019 Chartres Encampment achieved a 70 percent housing placement rate. System-wide two-year housing retention hit 90 percent (Coalition for the Homeless of Houston/Harris County, 2024). Field teams fully resolved 127 encampments between 2021 and 2023. 75 percent of engaged individuals accepted housing navigation.

Houston validates the by-name HMIS registry as the operational backbone of the Field Architecture. It validates pre-staged unit matching as the mechanism for achieving 58 to 70 percent acceptance rates in long-term encampment populations. It validates the Leaf Blower Effect: encampment clearance without simultaneous housing placement produces redistribution, not population reduction.

11.2. Finland: National Housing First Program

Finland executes at national scale concentrating efforts in Helsinki (Y-Foundation, 2023). Interventions converted legacy congregate shelters into permanent, scatter-site, and supported housing units. The Y-Foundation functioned as a non-profit institutional landlord buying, building, and leasing at scale. The system executed multi-stakeholder collaboration across national government, municipalities, and NGOs. Housing preceded all treatment conditions. The architecture requires no sobriety, employment, or behavioral requirements for entry.

Measured outcomes demonstrate a 68 percent reduction in long-term homelessness between 2008 and 2022. Helsinki achieved dramatic reductions in emergency shelter beds generating a corresponding

increase in permanent supported housing units. Documented cost-effectiveness proved savings in emergency health, law enforcement, and judicial costs exceeding program costs.

Finland validates the governing principle. Housing serves as the clinical precondition, not the reward. It validates the non-profit institutional landlord model as an analog to MDI's non-profit co-location partners. It validates scatter-site and supported housing as the two-track architecture matching MDI's ALMU and field architecture structure.

11.3. The French Gardien Model

France operates at national scale across thousands of high-rise social housing complexes (Harvard Graduate School of Design, 2023; Eindhoven University of Technology, 2024). The live-in Gardien occupies a ground-floor residential loge within the building and functions as social broker, maintenance coordinator, and informal conflict mediator. Intimate familiarity with individual residents allows recognition of behavioral baselines and deviations.

Social cohesion in large-scale Habitation à Loyer Modéré (HLM) complexes correlates with Gardien engagement strength. Buildings with reduced or consolidated Gardien presence show measurable social fragmentation. The Gardien validates the Pod Steward model demonstrating lived-in presence, social broker function, the distinction between security enforcement and recognition-based welfare response, and the one to 150 staffing ratio as the working scale for effective social cohesion maintenance (Dunbar, 1992).

11.4. EmPATH: Emergency Psychiatric Assessment Treatment and Healing

The model deploys non-coercive psychiatric emergency unit design in urban emergency departments across the United States (American College of Emergency Physicians, 2020). The architecture features an open-floor-plan unit with recliner chairs replacing institutional beds. It presents a residential aesthetic. It emphasizes rapid stabilization. It positions de-escalation as environmental design that structurally reduces the necessity for pharmacological first response.

Measured outcomes demonstrate significant reductions in psychiatric boarding times and involuntary commitment rates compared against standard emergency department psychiatric intake. The literature documents reduced use-of-force incidents.

EmPATH validates the Zone A Sally Port and Zone B Triage ground floor design philosophy. It validates the anti-institutional aesthetic requirement and the non-coercive de-escalation approach as clinically superior to security-first intake.

11.5. CIT: Crisis Intervention Team Memphis Model

A specialized law enforcement co-responder model remains active across hundreds of United States municipalities (CIT International, 2023). The protocol mandates 40-hour specialized training for law enforcement officers. The co-responder pairing embeds a mental health clinician with a CIT officer. The mechanism routes individuals into designated psychiatric emergency facilities, bypassing criminal jail. Verbal de-escalation functions as the primary tool.

Measured outcomes prove a 28 to 58 percent reduction in use-of-force incidents across documented municipal studies establishing an approximate 70 percent psychiatric diversion rate for eligible cases. The model works strictly sequentially addressing one individual precisely independent of others. It cannot manage simultaneous multi-actor escalation events. Kinetic separation via the Sally Port represents the architectural equivalent enabling single-actor CIT de-escalation at MDI.

CIT validates dual-role crisis node staffing in Zone A and Zone B. It establishes why multi-actor escalation requires architectural separation before clinical de-escalation can function.

11.6. Adaptive Reuse Precedents

Crosstown Concourse in Memphis converted a 1.5 million square foot former Sears warehouse into a mixed-use vertical community incorporating housing, non-profit offices, education, arts, and retail in a single vertical ecosystem (Crosstown Arts and Crosstown Concourse LLC, 2017). Mixed-use integration creates organic daily encounters across demographic groups. Ground-floor activation connects the building to the surrounding neighborhood. Crosstown validates the non-profit co-location model as architecture (Los Angeles City Planning, 2026a). Visible mixed-use activation and community permeability reduce the stigma attached to stabilization housing.

Le Corbusier's 1952 Unité d'Habitation in Marseille established the conceptual precedent that vertical density does not preclude community formation when common infrastructure is deliberately embedded in the residential fabric (Corbusier, 1952). 1,600 residents across 17 stories, with an internal commercial street, rooftop gymnasium, and nursery. The "streets in the air" concept prefigures the MDI pod-level sky garden and common kitchen placement logic.

12. The Prototype Verification Threshold

The MDI thesis is falsifiable. The Singular Prototype Threshold at One California Plaza defines eight binary verification metrics. All eight must pass before the first replication tower is acquired. Failure of any single metric halts network expansion and triggers architectural revision.

12.1. The Eight Verification Metrics

Table 9 specifies the eight binary thresholds constituting the Singular Prototype Threshold gate.

#	Metric	Threshold	Source
1	Encampment reduction in target zone	70 to 80 percent within 24 months of tower activation	LAHSA PIT methodology
2	Continuous housing retention	≥ 85 percent continuous residency at 24 months post-placement	HMIS registry tracking
3	Emergency service cost reduction	$\geq \$40,000$ per resident per year	County fiscal audit
4	Voluntary acceptance rate (Pipeline B)	≥ 55 percent at targeted warm offer	Field ACT team records
5	Legal lever Pipeline C augmentation	≥ 15 percent of total C-cohort housed via compelled pathways	CARE Court/AOT records
6	Capital recovery timeline	≤ 45 months from activation to breakeven	Stewardship Authority audit
7	Sustained Phase Zero clearance	≥ 70 percent maintaining ≥ 2 indicators across 60 to 90 day window	MHRC clinical records
8	System throughput integrity	Zero backlog formation exceeding 15 days for Phase Zero entry	MHRC flow regulator audit

Table 9. Singular Prototype Threshold: Eight Binary Verification Metrics.

Metric 7 is the causal verification metric: without it, a retention gain at Metric 2 cannot be attributed to the Metabolic Stabilization sequencing design versus any other variable in the pipeline. Metrics 1-6 independently verify population reduction, residential permanence, fiscal recovery, and the mechanical viability of both voluntary and compelled pathways. Metric 8 ensures that Phase Zero functions as a flow regulator rather than a systemic bottleneck. To prevent metric manipulation or documentation gaming, all threshold data are subject to an adversarial audit conducted by the Singular Prototype Oversight Committee.

12.2. The SPT Gate

No replication tower is acquired, converted, or activated until all eight metrics achieve threshold at One California Plaza. Because the Singular Prototype Threshold (SPT) operates as a binary gate, partial success cannot authorize partial expansion. The gate exists to prevent the political dynamics that have historically scaled unverified homelessness programs across California before outcome data existed. Bypassing the gate prior to successful metric validation requires a two-thirds supermajority override by the Stewardship Authority's governing board.

If the prototype fails any metric, the architectural revision process identifies the specific failure point, revises the design, and re-tests at the prototype site. The network does not expand on the basis of political pressure, media narrative, or projected outcomes. It expands on the basis of measured outcomes at a single verified site.

12.3. The Falsifiability Clause

The primary MDI thesis states that the physiological prerequisites for housing maintenance are the structural missing component explaining why Housing First achieves 80 to 90 percent housing retention in Finland and sub-40 percent in the United States (UCSF Benioff Homelessness and Housing Initiative, 2023). Metabolic Stabilization delivered through MHRC-licensed sub-acute clinical clusters before permanent placement provides these necessary prerequisites. The operational thesis is that a fully engineered pipeline delivering that prerequisite layer, combined with field architecture, a legal lever system, and a terminal infrastructure node operating simultaneously, can reduce chronic unsheltered homelessness in a targeted metropolitan zone by 70 to 80 percent within 24 months of tower activation, sustain retention at or above 85 percent, and recover capital investment within 45 months.

If One California Plaza, operating at full specification with all simultaneous operations active for 24 months, fails to achieve these thresholds, the thesis is falsified at both levels. The Metabolic Stabilization prerequisite layer does not produce the physiological clearance claimed. The pipeline engineering model does not produce the claimed population-level outcome. Failure invalidates the model. It proves the structural barriers exceed the framework's engineering capacity, requiring foundational revision.

Achieving the retention threshold alone cannot isolate the causal impact of Metabolic Stabilization from the architectural advantages of the ALMU environment or from potential selection effects associated with stabilization completion. To isolate this variable, the prototype mandates a nested Intent-

to-Treat Randomized Controlled Trial (RCT). A defined high-acuity cohort must be randomized at initial contact into two arms: Arm A receiving Phase Zero stabilization prior to placement, and Arm B receiving immediate placement into the identical ALMU environment under equivalent service conditions. All participants are analyzed according to initial assignment, and non-completion of stabilization in Arm A is counted as a negative outcome. Continuous housing retention is defined as uninterrupted residence without administrative reset or transfer masking. This primary causal thesis fails if Arm A does not achieve at least 70 percent absolute 12-month retention, or if the difference in 12-month continuous housing retention between arms falls below 15 percentage points or fails to achieve statistical significance under intent-to-treat analysis. While this 12-month RCT structurally isolates the causal mechanism of stabilization, SPT Metric 2 evaluates the durability of the full pipeline system at 24 months. Both tests must independently verify. Furthermore, to prevent gaming, the RCT requires a pre-operations protocol establishing an independent IRB randomization authority and an external data custodian. Finally, the entire operational model remains subject to the CMS interpretation risk regarding IMD facility clustering, which operates independently of clinical outcomes.

This clause is the difference between a policy proposal and an engineering specification. Policy proposals are evaluated on intention. Engineering specifications are evaluated on measured performance against stated thresholds.

13. Conclusion

The street homelessness crisis in Los Angeles persists because the system is sequenced wrong. Housing is offered to individuals who cannot maintain housing because the physiological prerequisites for tenancy have not been established. Finland proved that establishing those prerequisites before placement produces 80 to 90 percent 2-year housing retention. The United States produces sub-40 percent retention because it deploys housing without the prerequisite layer (UCSF Benioff Homelessness and Housing Initiative, 2023). 24 billion dollars in California expenditure and four decades of distributed shelter programming have not changed this sequencing error (California State Auditor, 2026; Y-Foundation, 2023; FEANTSA, 2022).

The Material Dignity Infrastructure resequences the pipeline. Phase Zero defines the comprehensive pre-admission structural sequence including the clinical stabilization layer that operates before any permanent housing placement occurs. This is the missing layer. It operates under existing California law, funded through existing Medi-Cal sub-acute billing at 50 percent federal financial participation, requiring a Section 1115 demonstration waiver anchored to the California CalAIM framework (Centers for Medicare & Medicaid Services, 2023) to resolve the Centers for Medicare and Medicaid Services interpretation risk. The Not In My Backyard attack surface is resolved by medical facility classification. The diagnostic protocol is audit-defensible (California Department of Health Care Services, 2024c).

Five distinct population pipelines receive five matched instruments. Pipeline A resolves through coordinated entry acceleration. Pipeline B resolves through a specific, immediate warm offer eliminating the rational refusal barriers of pet separation, partner separation, and possession loss. Pipeline C Calcified resolves through 12 to 24 months of ACT sustained presence combined with the legal lever system for individuals whose neurology precludes voluntary engagement. Pipeline C Riparian resolves through environmental abatement and simultaneous housing placement. The voluntary nomadic individuals in Pipeline D receive harm reduction without placement pressure. No legal instrument applies to Pipeline D. The Anti-Detention Covenant protects every resident's right of egress. All three intake pathways feed the same pipeline toward the same destination. Whether entering via Sally Port, open walk-in, or secondary clinical transfer, individuals route to a named room, a stored cart, a kenneled pet, and a keycard issued at the moment of willingness.

The population does not need to be cleared. It needs to be reduced substantially. Houston achieved 60 percent reduction using coordinated entry as its primary tool over nine years. MDI deploys

five simultaneous operations with a stronger toolset, a purpose-built inventory, and a metabolic stabilization layer Houston lacked. The 70 to 80 percent reduction target at the prototype zone is conservative relative to the comparative evidence base. County-wide reduction to that threshold requires 16 to 20 tower deployments across a five to ten year network timeline. The single prototype proves the pipeline. The network delivers the population-level outcome.

The efficiency dividend that follows is not fiscal alone. The six-layer dividend reaches every public system that currently manages the chronic street crisis as an unresolvable baseline. Emergency rooms recover psychiatric bed capacity. Law enforcement reallocates officer time from encampment management to other calls. Sanitation operations transition from crisis remediation to routine maintenance. The Financial District corridor around One California Plaza recovers commercial viability. Community hospitals serving the housed population are relieved of the behavioral health surge the unhoused population currently generates. Every front line worker experiences materially improved working conditions when the population they manage decreases by 70 to 80 percent. This dividend benefits outreach workers, ACT clinicians, ER nurses, LAPD officers, and sanitation crews. These workforce sustainability gains are not counted in the Efficiency Surplus. They are system-level dividends that accrue to every institution that touches the crisis.

The population that remains after substantial Pipeline C housing represents a qualitatively different street demographic that is smaller in total volume, less acutely destabilized, and significantly less visible as a public safety emergency. As the population shrinks, the behavioral contagion effect that typically amplifies crisis in dense encampment conditions diminishes. Consequently, the voluntary nomadic individuals in Pipeline D operate in a street environment no longer dominated by acute metabolic destabilization. They become more reachable by harm reduction services, face fewer risks from the secondary dangers of a crisis-dense encampment environment, and grow more likely to self-select into ground floor open access over time.

Because the street homelessness crisis is a sequencing failure, the Material Dignity Infrastructure reorganizes the operational timeline to establish physiological prerequisites before housing placement. This framework shifts the response from a policy intention to a falsifiable engineering specification. By defining clear verification thresholds, engineering a dedicated prerequisite layer, and specifying a rigid prototype model, the framework guarantees that the target population must diminish. This structural reorganization provides the exact mechanism by which that reduction occurs.

A. Operational Lexicon

The following lexicon defines all terms, acronyms, and operational concepts used throughout this manuscript. Entries are organized thematically. Part VIII provides a complete acronym quick-reference index.

Part I: Core MDI Framework Terms

Anti-Detention Covenant The civil liberty guarantee codified in the Stewardship Authority charter. Guarantees unrestricted egress for all residents at all times following admission. Governs voluntary residency only. Distinct from existing California legal hold authority (5150 WIC, LPS Conservatorship), which is statutory law predating MDI.

ALMU (Asset Limited Modular Unit) The precision-engineered 150 sq ft residential module within the MDI tower. Classified as a non-property interest. Includes independent wet bath, Murphy wall-bed, locking storage, induction cooktop, mini-refrigerator, and STC 65 acoustic door. Only the resident holds the key.

Automated-Human Operational Balance The governing cybernetic principle. Automation applied to the mechanical layer (safety telemetry, robotic maintenance, logistics). Human judgment applied to the relational layer (Pod Steward welfare checks, peer specialist engagement, crisis response). Total automation and total human management are both rejected.

Bulk Majority Argument The thesis that clearing 70 to 80 percent of the Pipeline C street population produces a population ecology shift, rendering the remaining voluntary nomadic Pipeline D population manageable via harm reduction rather than crisis management.

Dunbar Group (DG) An ITP-assigned social cluster within a Dunbar Pod. Sizing is individualized: 5 to 8 individuals for socially sensitive residents; up to 20 to 30 for those with higher social tolerance. DG assignment is revisable as the resident stabilizes. Smaller DG is a clinical match, not a lower-status outcome.

Dunbar Pod One of 13 independent residential partitions within the MDI tower. Approximately 154 units per pod, yielding 2,000 total. Four to five floors. Independent elevator loop. Time-slot-scheduled shared amenity access via RFID lockout. Derived from Dunbar's Number cognitive ceiling of approximately 150 for stable social relationships.

Efficiency Surplus The net annual public financial benefit per tower generated through municipal service cost elimination. Best case: 82 million dollars per year. Most probable: 52.5 million dollars per year. Worst case: 33 million dollars per year. Capital recovery timeline: 29–71 months. Six-layer dividend: ER throughput, law enforcement reallocation, sanitation, commercial district recovery, community hospital relief, and front line worker conditions.

Exoskeletal Identity Structure The functional role of a Pipeline C individual's accumulated hoard. Provides the physical boundary of a self-constructed world. Not random accumulation. Removing the hoard without replacing the identity structure triggers acute psychiatric destabilization. Intervention must be identity-preserving.

Tenancy Bridge Guarantee The contractual mechanism ensuring the Stewardship Contract remains active as a legal guarantor of tenure until a new private market lease is verified, signed, and funded. No resident transitions to the independent market without a confirmed landing.

Coordinated Unit Activation The simultaneous opening of all pre-matched tower units at One California Plaza. Transforms the housing offer from a generic referral into a specific, immediate, and irreversible commitment: named room, stored cart, kenneled pet, keycard issued at the moment of willingness.

Leaf Blower Effect The consequence of encampment clearance without simultaneous housing placement. Produces geographic redistribution, not population reduction. ACT engagement relationships are severed. Engagement timelines reset to zero.

MDI (Material Dignity Infrastructure) The end-to-end industrial pipeline for the engineered elimination of chronic street homelessness. Comprises Phase Zero (Metabolic Stabilization), Phase One (Field Architecture), Phase Two (Ground Floor Intake), Phase Three (Ontological Architecture), and five simultaneous operations.

Metabolic Stabilization (MS) Phase Zero of the MDI pipeline. The 30 to 90 day clinical stage delivering sub-acute metabolic, neurological, and nutritional stabilization before any permanent housing placement occurs. The prerequisite layer that Housing First requires but the United States system does not currently provide.

MS Unit The Metabolic Stabilization Unit. The clinical residential environment where Phase Zero stabilization occurs. Operated under MHRC licensure. Distinct from the ALMU residential track.

Nocturnal Access Protocol Automated throttling of discretionary common floor access between 0000 and 0500 hours. Normalizes institutional social rhythm to match market-rate residential buildings. Medical, mental health, and safety access remain unrestricted globally.

Ontological Permanence Architecture (Ontological Security) The physical and operational mechanisms providing structural predictability sufficient to eliminate the routine of being roused (Giddens, 1991; Laing, 1960). Core markers: space for daily routines, sense of privacy, control over environment, coherent sense of self. The ALMU delivers all four simultaneously.

Pod Steward A credentialed peer support specialist with lived experience of homelessness or psychiatric illness. One per Dunbar Pod of 154 individuals. Residential unit within the pod preferred. Primary function: recognition, not enforcement. Human welfare interpreter of life-safety telemetry. The relational layer of the Automated-Human Operational Balance.

Population Ecology Shift The qualitative change in the street environment produced by substantially housing Pipeline C. Smaller remaining population. Less acute destabilization. Less visible public safety emergency. Diminished behavioral contagion. Changes Pipeline D management from crisis response to harm reduction.

Rapid Transition Protocol The mechanism replacing eviction. Transfers a resident to a different pod or tower within 24 hours when behavior creates physical danger for pod members and de-escalation has failed. Medical necessity review precedes transfer. Resident retains Stewardship Contract. No resident becomes homeless again through behavioral crisis.

Sally Port (Pathway 1) The formal, pre-matched intake pathway. Mobile crisis outreach and ACT team-facilitated warm offer to a named individual on the by-name HMIS registry. Unit, pod, and floor are pre-matched before the offer conversation begins. Highest-quality intake: the individual is known, pre-matched, and entering at a chosen moment of willingness.

Singular Prototype Threshold (SPT) The eight binary verification metrics that must all pass at One California Plaza before any replication tower is acquired. A binary gate. Partial success does not authorize partial expansion. Metric 7 (Phase Zero clearance rate) is the causal test of the primary academic thesis.

Sovereign Acquisition via Receivership The Stewardship Authority's mechanism for acquiring distressed commercial towers through court-supervised bankruptcy proceedings at verified floor valuations (typically 120 dollars per square foot; 70 to 85 percent below peak CRE valuations).

STC 65 Phase Zero Module The physical module of each MHRC cluster. A 65-decibel sound enclosure designated as a state-owned clinical environment. Not a residential dwelling. Not a shelter. A therapeutic milieu unit analogous to a hospital step-down bed.

Stewardship Authority The state-chartered quasi-governmental entity modeled on the Tennessee Valley Authority. Possesses delegated state police power and independent bonding capacity. Functions as MDI operator, CARE Court petitioner, real estate acquirer, and outcome verifier. Addresses the coordination failure that paralyzes existing Los Angeles homelessness governance.

Stewardship Contract The legal instrument governing residential tenure within the tower. Classified as a non-property interest. Resident holds occupancy, privacy, and non-interference rights equivalent to a residential lease without triggering housing court eviction jurisdiction.

Hygiene Intake Kit The gift offered to Diogenes-pattern individuals at hygiene intake. Framed as addition, not replacement. Staff bag and tag soiled clothing pending individual consent for disposal.

Resident Civic Engagement Pathway Voluntary participation tracks transitioning residents from passive service recipients to active stakeholders. Includes Garden Stewardship, Robotic Swarm Supervision, Community Kitchen Coordination, and Digital Access Center Support. Earns Stewardship Status, a social recognition category within the pod.

Three Ps Infrastructure The structural elimination of three primary shelter refusal barriers: Pets (on-site facility, veterinary node, dog run); Partners (co-location within same Dunbar Pod); Possessions (photographed, GPS-tagged, RFID-accessible secure storage).

Warm Offer A specific, immediate, irreversible housing offer made at the moment of the individual's first window of willingness following an acute crisis event. Pre-confirmed unit, kenneled pet, stored possessions, completed intake documentation. Eliminates bureaucratic delay and every rational refusal barrier simultaneously.

Part II: Population Pipelines

Pathway 3 (Secondary Clinical Transfer) Individual transferred following field contact, non-criminal law enforcement determination, or ER discharge to a lower-acuity setting. Uses the rear-wall Zone A entrance, physically separated from the open commons. Prevents crisis contagion in the resource center population.

Pipeline A The near-homeless and voluntarily transitioning. No major psychiatric comorbidity. Engages voluntarily with the Coordinated Entry System. Rapid placement track. CES is the existing instrument; MDI accelerates it by creating surplus inventory.

Pipeline B The encamped but engageable. Long-term encampment with intact insight. Refusal directed at congregate shelter barriers (pets, partners, possessions), not housing itself. Warm offer at acute crisis window. 58 to 70 percent acceptance rate with a specific, immediate offer.

Pipeline C The behaviorally calcified chronic. Long Duration of Untreated Psychosis (5 to 15 years). Active schizophrenia spectrum disorder. Requires ACT sustained presence 12 to 24 months plus the legal lever system. Primary driver of public safety concern and street visibility.

Pipeline C Riparian Sub-Variant The hidden riparian population. Mobile, terrain-adaptive, encamped in environmentally sensitive sites (LA River, Ballona Creek, storm drains). Estimated 20 to 30 percent above PIT count Pipeline C figure due to systematic undercount. Requires environmental enforcement track (field mapping + CWA abatement) paired with simultaneous warm offer.

Pipeline D The voluntary nomadic. Not metabolically destabilized in the Pipeline C clinical sense. Organized life around outdoor community, informal economy, and deliberate rejection of institutional structures. Instrument: harm reduction without housing requirement. No legal instrument applies. Clearance of Pipeline C changes the street environment sufficiently that Pipeline D becomes manageable via harm reduction alone.

Part III: Clinical Terms

Anosognosia A neurological incapacity to perceive one's own psychiatric condition. Up to 50 percent of schizophrenia-spectrum individuals present with it. The frontal lobe damage producing psychosis simultaneously destroys the self-monitoring faculty. Outreach strategies premised on persuasion target a faculty already destroyed by the disease.

Behavioral Calcification The progressive rigidity of survival routines and threat-perception systems produced by years of untreated psychosis and street-level chronic stress. Key barrier distinguishing Pipeline C from Pipeline B. Duration directly correlates with intervention resistance.

Day-30 Functional Indicators The five ITP measurable goals used as the Phase Zero clearance threshold: (1) sustained hygiene maintenance without prompting; (2) six or more hours sleep without pharmacological support; (3) orientation to person, place, and date across three consecutive days; (4) one structured group activity per day; (5) one independent ADL task per day. Progress on two or more equals clearance.

Diogenes Syndrome Extreme self-neglect, domestic squalor, social withdrawal, and adamant refusal of help. The accumulated hoard functions as an Exoskeletal Identity Structure. Intervention must be identity-preserving. Hoard removal without identity structure replacement triggers acute destabilization.

DSM-5 (Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition) The APA diagnostic classification system governing all provisional and finalized diagnoses throughout the Phase Zero Three-Phase Diagnostic Protocol.

DUP (Duration of Untreated Psychosis) The accumulation of years of untreated psychosis producing measurable progressive brain changes: increasing cognitive rigidity, treatment resistance, and functional decline. Pipeline C typically presents 5 to 15 years DUP. Standard clinical timescales do not apply to this cohort.

Hyponatremia Potentially fatal cerebral edema produced by rapid water loading in severely dehydrated individuals with depleted sodium. Life-safety constraint on the rehydration protocol: WHO-formulated electrolyte solution is required before water. Governs the robotic kitchen and pharmacy protocols.

Refeeding Syndrome Life-threatening electrolyte shifts (cardiac arrhythmia, neurological complications) produced by rapid carbohydrate reintroduction in severely malnourished individuals. Clinical triage layer required at intake. BMI below 16, visible severe wasting, or reported days without food triggers the Gradual Reintroduction Protocol before open cafeteria access.

Three-Phase Diagnostic Protocol Phase 1 (Hours 0 to 72): Provisional DSM-5 diagnosis by LPHA; billable as Mental Health Services Assessment. Phase 2 (Days 1 to 14): Collateral gathering, daily progress notes, sustained billing trail. Phase 3 (By Day 30): Finalized DSM-5 diagnosis signed by LPHA; ITP signed; full Medi-Cal billing authorized.

Wernicke Encephalopathy Neurological emergency produced by Thiamine (Vitamin B1) deficiency during nutritional refeeding. Clinical contraindication to rapid carbohydrate reintroduction in severely malnourished arrivals. Governed by the Refeeding Syndrome clinical triage layer.

Part IV: Legal and Regulatory Terms

5150 Hold California Welfare and Institutions Code Section 5150. Authorizes a 72-hour involuntary psychiatric hold for individuals posing danger to self, danger to others, or meeting grave disability criteria. Predates MDI. Provides the acute triage authority during the crisis window. Not an MDI-created authority.

AOT (Assisted Outpatient Treatment) Court-ordered outpatient treatment for individuals with documented history of treatment non-compliance and repeated psychiatric hospitalizations, incarcerations, or threats/acts of violence. Governed by Laura's Law (2002). Mandates medication compliance and outpatient engagement. Non-compliance triggers court review and potential LPS escalation.

ARO (Adaptive Reuse Ordinance) The Los Angeles ordinance (February 2026 expansion) eliminating minimum unit size requirements and public hearing mandates for qualifying commercial-to-residential conversions. The MDI ALMU at 150 sq ft qualifies under the expanded ARO.

CARE Court Community Assistance, Recovery, and Empowerment Act (2022). California court mechanism for adults with schizophrenia spectrum or other psychotic disorders. Non-custodial. Establishes a 12 to 24 month court-monitored CARE Agreement including housing, medication, and wraparound services. The Stewardship Authority petitions as a qualifying entity using by-name HMIS registry documentation.

CWA (Clean Water Act, 1972) Federal statute providing the legal basis for environmental abatement of riparian encampments. Fecal coliform contamination exceeding NPDES permit standards and biological waste infiltration into protected habitat constitute documented violations providing abatement authority independent of housing law.

Federal Supremacy Leasing Legal mechanism invoking federal preemption of conflicting municipal regulations for tower space occupied by federal agencies or federally funded programs. Creates a secondary legal shield for federal service co-location floors against municipal obstruction.

Grants Pass Framework *City of Grants Pass v. Johnson* (2024). The Supreme Court ruling governing municipal authority over encampment clearance. Riparian abatement is explicitly outside this framework, as the legal basis is environmental protection (CWA, Army Corps jurisdiction), not anti-camping ordinances.

Home Rule Stabilization Ordinance State ordinance pre-empting conflicting local land use regulations for qualifying projects. Codifies state housing law supremacy into automatic by-right conversion authority for commercial structures acquired through the Sovereign Acquisition process.

IMD Exclusion (Institution for Mental Disease) The federal Medicaid rule excluding federal financial participation for mental health facilities with more than 16 beds. Resolved architecturally by the sub-16-unit MHRC cluster structure: each cluster falls below the threshold and files independently. No waiver required.

LPS Conservatorship (Lanterman-Petris-Short Act) Highest-restriction legal instrument. Court appoints Public Guardian or designated conservator for individuals deemed gravely disabled and unable to provide for food, clothing, or shelter due to mental health disorder or severe substance use disorder (SB 43 expansion, 2023). Conservator may consent to housing placement and medical treatment. Subject to periodic court review.

Medical Classification Waiver DHCS petition enabling ALMU residential units to be classified as community-based residential treatment for Medi-Cal billing purposes. Preserves residential character while enabling clinical billing that funds the field architecture.

NPDES (National Pollutant Discharge Elimination System) Clean Water Act permitting system. Fecal coliform contamination of LA waterways from encampments exceeds NPDES permit standards, constituting documented federal regulatory violations and the legal basis for environmental abatement proceedings.

SB 330 State law vesting zoning entitlements at the date of application for qualifying projects. Prevents subsequent zoning amendments from altering the entitlement after the Stewardship Authority files a conversion application.

Stewardship Agreement The resident's contractual instrument under the Stewardship Contract. Grants occupancy, privacy, and non-interference rights. Classified as a non-property interest to bypass housing court jurisdiction while preserving residential dignity protections.

TVA Model (Tennessee Valley Authority) The template for the Stewardship Authority. Quasi-governmental, state-chartered, operationally autonomous from municipal government. Consolidates field operations, legal petitioning, real estate acquisition, management, and outcome verification under a single mandate-limited entity.

WIC (Welfare and Institutions Code) California statute. Section 5150 provides the 72-hour involuntary psychiatric hold authority applied in the Three-Gate Triage Decision Architecture as the governing authority for the Emergency Room pathway.

Part V: Operational and Field Architecture Terms

ACT (Assertive Community Treatment) Evidence-based intensive outreach model. Team composition: mobile psychiatrist (shared), RN, licensed social worker, outreach case manager, peer support specialist, substance use counselor, team coordinator (7-8 FTE). Caseload ratio: 80-120 Pipeline C clients. Medi-Cal-billable in California at 150to190 per day per enrolled client.

By-Name HMIS Registry The operational backbone of the Field Architecture. Every individual in the target zone registered by name in an HMIS instance managed by the Stewardship Authority. Every housing unit pre-matched to a named individual before tower opening. The warm offer is specific because the registry preparation was specific.

CES (Coordinated Entry System) The existing Los Angeles coordinated intake and assessment system for homeless services. Pipeline A individuals engage voluntarily through CES. MDI accelerates CES by creating surplus inventory.

CIT (Crisis Intervention Team) Specialized law enforcement co-responder model. 40-hour specialized training. Mental health clinician embedded with CIT officers. Routes individuals directly into designated psychiatric emergency facilities, bypassing criminal jail. Validates dual-role crisis node staffing in Zone A and Zone B.

CPTED (Crime Prevention Through Environmental Design) Architectural alternative to camera-dense digital surveillance. Designs spaces enabling residents to naturally observe common areas as part of daily movement. Glass-partitioned common rooms, high-visibility stairwells, digital access center, and pod kitchen at circulation chokepoints.

EmPATH (Emergency Psychiatric Assessment Treatment and Healing) Non-coercive psychiatric emergency unit design model. Open floor plan, recliner chairs, residential aesthetic, rapid stabilization, de-escalation as environmental design first response. Validates Zone A and Zone B ground floor design philosophy.

Outreach Workers MDI field assessment personnel. Jointly with ACT teams and law enforcement, apply the Three-Gate Triage Decision Architecture to every crisis presentation.

HMIS (Homeless Management Information System) HUD-required data system for tracking homeless individuals and services. The MDI Stewardship Authority operates a proprietary HMIS instance as the by-name registry backbone.

LAHSA (Los Angeles Homeless Services Authority) The joint authority of the City and County of Los Angeles responsible for coordinating and implementing homeless services. Administers the annual Point-in-Time count. Source data for MDI field architecture sizing.

LARWQCB (Los Angeles Regional Water Quality Control Board) State regulatory body governing water quality in the Los Angeles region. Enforcement records document Clean Water Act violations produced by riparian encampments. Provides the legal basis for environmental abatement independent of housing law.

PIT Count (Point-in-Time Count) The annual one-night survey of the homeless population mandated by HUD. Systematically undercounts the most hidden encampments, particularly Pipeline C Riparian individuals. 2025 Greater Los Angeles PIT: approximately 75,312 total homeless, 58 percent unsheltered.

Part VI: Architectural and Physical Design Terms

Residential Data Minimization Protocol The only data exported from a residential ALMU to the central cybernetic grid is a binary safe/not-safe occupancy indicator. All other environmental data processed locally and purged within 24 hours. Structurally precludes behavioral profiling and the Panopticon Effect.

Biophilic Infrastructure Two-tier system. Tier 1: 200-300 sq ft planted pod-level alcove on every 4-5 floors, adjacent to pod common kitchen. Tier 2: Tower-level destination sanctuary every 7 floors with hydroponic garden, oxygenated lounge, rooftop greenspace. Reduces cortisol, improves mood regulation, decreases agitation incidents.

Real-time Unit Availability Display Live display of unit and resource availability across the network. Deployed in the ground floor Resource Center and Transportation Asset Management system. Neutralizes scarcity-driven anxiety before it becomes behavioral agitation.

Digital Access Center Low-distraction, high-bandwidth vocational and educational access hub. Gigabit internet, hardware lending, vocational training, legal self-defense resources (benefits enrollment, ID restoration, credit rehabilitation), gig-work platform access.

HEPA (High Efficiency Particulate Air) Air filtration standard. Clinical-grade HEPA scrubbing deployed in ALMU units and acuity-segmented hygiene modules. Air quality exceeds average Los Angeles apartment standards.

HLM (Habitation à Loyer Modéré) French social housing designation. Large-scale high-rise social housing complexes where the Gardien/Pod Steward model was developed and validated.

HVAC (Heating, Ventilation, and Air Conditioning) Individual HVAC control within a 68 to 78 degree operating band in each ALMU. Tower-level Tier 2 biophilic sanctuary operates at 50 percent of original HVAC design load surplus, producing elevated oxygen concentration.

MAP (Managed Alcohol Program) Medically supervised alcohol management facility for individuals in active, functionally incapacitating heavy substance dependence. Pre-ALMU track for this sub-population. Post-MAP stabilization, transfer to the ALMU track follows.

MHRC (Mental Health Rehabilitation Center) A 24-hour DHCS-licensed facility classification for adults with mental disorders providing intensive rehabilitative services for sub-acute stabilization, skill-building, and independence. The MDI Phase Zero cluster model. Sub-16-unit threshold resolves the IMD Exclusion architecturally.

MTBF (Mean Time Between Failures) Equipment reliability standard. Pressurized sewage ejectors at One California Plaza carry a 50,000-hour MTBF mandate.

RFID (Radio Frequency Identification) Electronic identification token system. Used for pod boundary enforcement on shared amenity floors via time-slot scheduling, secure possession storage access, and bicycle transportation asset tracking.

STC (Sound Transmission Class) Measure of sound attenuation performance. STC 55 is the biological floor (prevents hearing damage). STC 65 is the MDI ontological threshold, which reduces ambient sound transfer to clinical quiet-room standards, neutralizing hypervigilance triggers in trauma-impacted residents.

UV-C (Ultraviolet-C) Germicidal ultraviolet light wavelength. UV-C scrubbing deployed in acuity-segmented hygiene modules and ALMU HVAC systems for clinical-grade disinfection.

VOC (Volatile Organic Compound) Airborne chemical compounds released by building materials. MDI biophilic node species selected for low-VOC properties, drought tolerance, and sensory appropriateness.

Part VII: Fiscal Terms

CRE (Commercial Real Estate) The distressed commercial real estate providing the MDI acquisition target. One California Plaza: 120 dollars per square foot floor valuation (approximate 80 percent below peak CRE valuations). The 16 million square foot network conversion opportunity across the distressed downtown corridor.

DHCS (Department of Health Care Services) California state agency. Licenses MHRC facilities via Form 1813. Governs Medi-Cal specialty mental health billing standards. MDI petitions DHCS for the Medical Classification Waiver.

FMAP (Federal Medical Assistance Percentage) The federal cost-sharing rate for Medi-Cal expenditures. Approximately 50 percent federal financial participation on MHRC sub-acute specialty mental health billing. The load-bearing fiscal base of Phase Zero.

Four-Layer Funding Stack Layer 0: MHRC Medi-Cal sub-acute billing (Phase Zero). Layer 1: ACT Medi-Cal billing (clinical services at 150 to 190 dollars per day per enrolled client). Layer 2: Measure Alpha and MHSA Full Service Partnership (field architecture, Pod Stewards, residential management). Layer 3: Efficiency Surplus (capital recovery through municipal service cost elimination). No layer depends on speculative administrative approval for core viability.

Measure Alpha Los Angeles County measure generating 843 million dollars annually for homelessness services. The MDI Stewardship Authority operationalizes this funding by providing a verifiable infrastructure destination with measurable per-resident outcome verification.

MHSA (Mental Health Services Act) California state act generating approximately 3.5 billion dollars statewide annually. Los Angeles County allocation: approximately 400 million dollars per year. ACT teams for the chronically homeless unsheltered are among the highest priority eligible expenditures under the Full Service Partnership category.

SAMHSA (Substance Abuse and Mental Health Services Administration) Federal agency funding ACT implementation through competitive grants of \$500K–\$2M per program.

Part VIII: Acronym Quick-Reference Index

Acronym	Expansion	Acronym	Expansion
ACT	Assertive Community Treatment	ITP	Individualized Treatment Plan
ALMU	Asset Limited Modular Unit	LAHSA	Los Angeles Homeless Services Authority
AOT	Assisted Outpatient Treatment	LARWQCB	LA Regional Water Quality Control Board
ARO	Adaptive Reuse Ordinance	LPS	Lanterman-Petris-Short Act
BMI	Body Mass Index	LPHA	Licensed Practitioner of the Healing Arts
CARE Court	Community Assistance, Recovery, and Empowerment Act Court	MAP	Managed Alcohol Program
CES	Coordinated Entry System	MDI	Material Dignity Infrastructure
CIT	Crisis Intervention Team	MHRC	Mental Health Rehabilitation Center
CPTED	Crime Prevention Through Environmental Design	MHSA	Mental Health Services Act
CRE	Commercial Real Estate	MS	Metabolic Stabilization
CWA	Clean Water Act	MTBF	Mean Time Between Failures
DG	Dunbar Group	NIMBY	Not In My Backyard
DHCS	Department of Health Care Services	NPDES	National Pollutant Discharge Elimination System
DSM-5	Diagnostic and Statistical Manual, 5th Ed.	PIT	Point-in-Time (Count)
DUP	Duration of Untreated Psychosis	REM	Rapid Eye Movement
EmPATH	Emergency Psychiatric Assessment Treatment and Healing	RFID	Radio Frequency Identification
FMAP	Federal Medical Assistance Percentage	SAMHSA	Substance Abuse and Mental Health Services Administration
		SPT	Singular Prototype Threshold
FTE	Full-Time Equivalent	STC 65	Sound Transmission Class 65 Module
HEPA	High Efficiency Particulate Air	STC	Sound Transmission Class
HLM	Habitation à Loyer Modéré	TVA	Tennessee Valley Authority
HMIS	Homeless Management Information System	UV-C	Ultraviolet-C
HVAC	Heating, Ventilation, and Air Conditioning	VOC	Volatile Organic Compound
IMD	Institution for Mental Disease	WIC	Welfare and Institutions Code

References

- Allwork Space (2026). Los Angeles clears path to turn offices into homes. Urban Market Reports, Allwork Space. Reports on the February 2026 Los Angeles Adaptive Reuse Ordinance expansion authorizing streamlined by-right conversion of qualified commercial assets to residential use.
- American College of Emergency Physicians (2020). The EmPATH unit model: Emergency psychiatric assessment, treatment, and healing. Technical report, American College of Emergency Physicians. Documents the non-coercive psychiatric emergency unit design: open floor plan, recliner chairs, residential aesthetic, rapid stabilization emphasis; significant reductions in psychiatric boarding times, involuntary commitment rates, and use-of-force incidents.
- California Department of Health Care Services (2024a). Assertive community treatment (ACT) medical billing guidelines. Technical report, California Department of Health Care Services. Establishes ACT as a Medi-Cal-billable service under the Community Mental Health Services framework; defines the one hundred fifty to one hundred ninety dollar per day per enrolled client reimbursement rate; documents eighty-five to ninety percent Medi-Cal eligibility among the chronically homeless population.
- California Department of Health Care Services (2024b). Full Service Partnership and ACT Billing Realization Metrics. Technical report, DHCS Behavioral Health Division. Establishing sixty percent historical utilization bounds for street-based mental health services.
- California Department of Health Care Services (2024c). Mental health rehabilitation center (MHRC) licensing and certification standards. Technical report, California Department of Health Care Services. Defines MHRC as a 24-hour DHCS-licensed facility for adults with mental disorders providing intensive rehabilitative services; establishes DHCS Form 1813 licensure pathway, specialty mental health Medi-Cal billing eligibility, and program standard requirements for sub-acute stabilization; applicable to Phase Zero Metabolic Stabilization cluster architecture under the MDI pipeline.
- California Department of Housing and Community Development (2024). Legal frameworks for single room occupancy and residential hotel tenancy exemptions. HCD Policy Brief. Details the legal precedent allowing specific high-support residential configurations to operate under licensure exemptions avoiding standard housing court eviction jurisdiction.

California Mental Health Services Oversight and Accountability Commission (2024). Mental health services act (Proposition 63) annual revenue and expenditure report. Technical report, MHSOAC. Documents approximately three point five billion dollars statewide annual revenue from the one-percent income tax surcharge; Los Angeles County allocation approximately four hundred million dollars per year; ACT for the chronically homeless is a highest-priority eligible expenditure under the Full Service Partnership category.

California State Auditor (2026). Homelessness in California: The state's approach has not ensured that dollar-level spending has achieved meaningful outcomes. California State Auditor Report. Documents approximately twenty-four billion dollars in state homelessness expenditures from fiscal year 2019 through 2026; identifies accountability gaps and fragmented program delivery as primary efficiency failures.

California State Legislature (2002). Assembly bill 1421: Assisted outpatient treatment (Laura's Law). California Legislature, 2001–2002 Session. Authorizes court-ordered outpatient psychiatric treatment for individuals with documented histories of treatment non-compliance, hospitalization, incarceration, or threats of violence.

California State Legislature (2022). Senate bill 1338: Community assistance, recovery, and empowerment (CARE) act. California Legislature, 2021–2022 Session. Establishes CARE Court proceedings for individuals with schizophrenia spectrum and other psychotic disorders; creates a court-supervised community treatment framework across all fifty-eight California counties.

California State Legislature (2023). Senate bill 43: Behavioral health. California Legislature, 2023–2024 Session. Expands the definition of grave disability under the Lanterman-Petris-Short Act to include individuals unable to provide for personal safety or necessary medical care due to severe substance use disorder; broadens conservatorship eligibility.

California State Legislature (2024a). Assembly bill 1976: CARE court implementation. California Legislature, 2023–2024 Session. Appropriates fourteen point seven million dollars statewide for CARE Court implementation; defines eligible expenditures for petitioning entities and county behavioral health departments.

California State Legislature (2024b). Senate bill 14: Community assistance, recovery, and empowerment (care) act expansion. Chapter 484, Statutes of 2024. Codifies the expansion of the CARE Court target population to include bipolar disorder with psychotic features, effective early 2026.

- CalMatters (2026). California homelessness funding 2026: The accountability standard. Public Policy Analytics, CalMatters. Analyzes the fifty-thousand-dollar annual per-person municipal service cost baseline for the chronically unhoused in California; documents accountability requirements for state and county homelessness expenditures.
- Centers for Medicare & Medicaid Services (2017). State medicaid director letter #17-003: Strategies to address the opioid epidemic. Technical report, Department of Health and Human Services. Details CMS interpretation of the IMD Exclusion, specifically addressing the facility clustering and campus integration guidance that presents legal risk for coordinated multi-cluster operations.
- Centers for Medicare & Medicaid Services (2023). CalAIM Section 1115 Demonstration Waiver: Standard Terms and Conditions. Technical report, U.S. Department of Health and Human Services. Authorizing innovative service delivery and sub-acute clustering under Medi-Cal reform.
- CIT International (2023). The memphis model: Crisis intervention team core elements. Technical report, CIT International. Documents the forty-hour specialized law enforcement training program, co-responder pairing with mental health clinicians, twenty-eight to fifty-eight percent reduction in use-of-force incidents, and approximately seventy percent psychiatric diversion rate.
- Coalition for the Homeless of Houston/Harris County (2024). The way home: Annual report 2024. Technical report, Coalition for the Homeless of Houston/Harris County. Documents sixty percent reduction in overall homelessness 2011–2020; fifty-eight to seventy percent housing placement rates at targeted encampment clearance events; ninety percent two-year housing retention system-wide; one hundred twenty-seven encampments fully resolved 2021–2023.
- Corbusier, L. (1952). *Unité d'Habitation, Marseille*. Architectural Press. Three hundred thirty-seven apartments housing sixteen hundred residents across seventeen stories; internal commercial street, rooftop gymnasium, and communal infrastructure embedded in the residential fabric; conceptual predecessor to the Dunbar Pod sky-garden and circulation loop design.
- Council on Tall Buildings and Urban Habitat (2024). Social cohesion in high-rise residential settings. Technical report, Council on Tall Buildings and Urban Habitat. Documents correlation between subgroup identifiability, shared transitional spaces, and social cohesion in high-rise residential environments; identifies anonymity as the single most reliable predictor of inter-resident conflict.
- Crosstown Arts and Crosstown Concourse LLC (2017). Crosstown concourse: Adaptive reuse of the former Sears distribution center, Memphis. Project Documentation. One point five million square foot adaptive reuse of a former Sears warehouse into a mixed-use vertical community incorporating housing, non-profit offices, education, arts, and retail; validates the four-to-one civilian-to-resident integration ratio.

- Dunbar, R. I. M. (1992). *Neocortex Size as a Constraint on Group Size in Primates*, volume 22. Journal of Human Evolution. Establishes the cognitive ceiling of approximately one hundred fifty for stable, trust-based social relationships; the theoretical basis for the Dunbar Pod partitioning framework.
- Eindhoven University of Technology (2024). The gardien effect: Live-in caretakers and social fragmentation in large-scale social housing. Technical report, TU Eindhoven, Department of the Built Environment. Documents measurable social fragmentation in French HLM social housing complexes following reduction or consolidation of live-in Gardien presence; validates the one-to-one-hundred-fifty staffing ratio.
- Environmental Protection Agency (2023). NPDES Compliance and Riparian Encampment Abatement Protocols. Technical report, EPA Office of Water. Establishing public health and environmental violation thresholds for riparian clearance.
- Estrada, I. (2026). Los Angeles approves conversion of empty offices into thousands of housing units. Real Estate Policy Briefs. Documents the municipal approval framework for commercial to residential conversion; identifies specific distressed asset categories eligible for authority acquisition.
- FEANTSA (2022). Finland's housing first policy: National program and international implications. Technical report, European Federation of National Organisations Working with the Homeless. Documents the Finnish Housing First national program; identifies the integrated national welfare infrastructure as the mechanism enabling unconditional housing placement with durable retention outcomes; establishes the metabolic stabilization gap as the transfer-failure variable for United States replication.
- Frontiers in Psychology Editorial Board (2024). Acoustic environment and psychological well-being in residential settings. *Frontiers in Psychology*, 15. Documents correlation between high-performance acoustic isolation (STC 60 and above) and reduced cortisol levels, improved sleep quality, and fostering of autonomy in residential occupants.
- GB Acoustics (2024). Sound transmission class performance standards for residential applications. Technical report, GB Acoustics. Technical specification document establishing that STC 65 renders loud speech inaudible in adjacent residential spaces; STC 65 approaches clinical quiet-room standards.
- Giddens, A. (1991). *Modernity and Self-Identity: Self and Society in the Late Modern Age*. Stanford University Press. Defines ontological security as the confidence that most human beings have in the continuity of their self-identity and in the constancy of their social and material environments; foundational to the MDI ALMU design thesis.
- GlobeSt (2025). Los Angeles aims to put housing on vacant lots. Commercial Real Estate Intelligence, GlobeSt. Documents the fifty million square foot vacant commercial inventory, the \$120 per square foot distressed floor valuation, and the city's policy framework for residential conversion.

- Grand Rising Behavioral Health (2024). Biophilic design in psychiatric and high-density residential settings. Technical report, Grand Rising Behavioral Health. Documents measurable cortisol reduction, improved mood regulation, and decreased agitation incidents in residential environments incorporating biophilic elements; validates passive exposure through daily movement patterns.
- Harvard Graduate School of Design (2023). The french social housing caretaker model: Implications for high-density residential management. *Harvard Design Studies*. Analyzes the Gardien model in French Habitation a Loyer Modere complexes; identifies the social broker function as distinct from security enforcement; documents correlation between Gardien engagement strength and resident-reported community cohesion.
- Holland and Knight (2025). California courts reemphasize that state housing laws prevail. Legal Analytics, Holland and Knight LLP. Documents judicial confirmation that state housing law supersedes conflicting local ordinances; establishes legal precedent for the Municipal Direct Bypass mandate under the Home Rule Stabilization Ordinance framework.
- IMEG Corp (2024). Biophilic infrastructure integration in high-density residential design. Technical report, IMEG Corp. Engineering specification for biophilic node integration in high-rise residential conversion; automated irrigation, species selection, and HVAC interaction parameters.
- Laing, R. D. (1960). *The Divided Self: An Existential Study in Sanity and Madness*. Penguin Books. Establishes the relationship between environmental predictability and psychotic fragmentation; identifies ontological insecurity as a precondition for schizoid withdrawal.
- Larimer, M. E. et al. (2009). Health care and public service use and costs before and after provision of housing for chronically homeless persons with severe alcohol problems. *Journal of the American Medical Association*, 301(13). Documents Seattle's 1811 Eastlake Housing First program for chronically homeless individuals with severe alcohol dependence; median cost savings of fifty-three percent per resident in publicly funded health care and services.
- Los Angeles City Planning (2026a). Adaptive reuse ordinance (February 2026 update). Technical report, Los Angeles City Planning Department. Defines by-right conversion eligibility for buildings at least fifteen years old; removes minimum unit size requirements and public hearing mandates for qualifying adaptive reuse projects.
- Los Angeles City Planning (2026b). Housing crisis act (SB 330) implementation. Technical report, Los Angeles City Planning Department. Implementation documentation for the Housing Crisis Act (SB 330); establishes anti-demolition protections, density preservation requirements, and expedited approval pathways for qualifying residential conversion projects.

- Los Angeles County Chief Executive Office (2025). Fiscal year 2026–27 Measure Alpha spending plan. Technical report, Los Angeles County Chief Executive Office. Eight hundred and forty-three million dollar approved spending plan; mandates one point two five percent accountability and data allocation generating thirteen point three eight million dollars for telemetric monitoring and outcome verification.
- Los Angeles County Department of Public Health (2026). Homeless mortality report 2026. Technical report, Los Angeles County Department of Public Health. Documents mortality rates, cause-specific death data, and life expectancy differentials for the unhoused population in Los Angeles County, fiscal year 2026.
- Los Angeles County Homeless Initiative (2025). Fiscal year 2025–26 approved funding recommendations. Technical report, Los Angeles County Homeless Initiative. Documents approved county funding streams for homelessness stabilization services, outcome-based allocation requirements, and Measure Alpha integration procedures for fiscal year 2025–26.
- Los Angeles Homeless Services Authority (2025). 2025 greater Los Angeles homeless count results. Technical report, Los Angeles Homeless Services Authority. Point-in-time count results for the Greater Los Angeles Continuum of Care; primary source for aggregate unhoused population estimates and demographic breakdown, 2025.
- Mehanna, H. M., Moledina, J., and Travis, J. (2008). Refeeding syndrome: What it is, and how to prevent and treat it. *British Medical Journal*, 336:1495–1498. Clinical guidelines for identifying and managing refeeding syndrome risk in severely malnourished individuals; documents the electrolyte shift mechanism and cardiac arrhythmia risk from rapid carbohydrate reintroduction.
- Mobile Loaves and Fishes (2024). Community first! village: Annual impact report 2024. Project Documentation, Mobile Loaves and Fishes. Documents eighty-five to eighty-eight percent housing retention rate for chronically unhoused individuals meeting HUD chronic homelessness definition; operating subsidy approximately twenty-five thousand dollars per resident annually; residents earned 1.5 million dollars collectively in 2023 through on-site Community Works program; validates the MDI retention target from United States operational data.
- NYU Marron Institute of Urban Management (2024). Smart home sensors in low-income housing: Privacy, autonomy, and the panopticon effect. *NYU Marron Institute Working Papers*. Documents behavioral alteration, performance stress, and erosion of home-as-sanctuary among residents subject to smart home sensor deployment in subsidized housing.

- Podymow, T., Turnbull, J., Coyle, D., Yetisir, E., and Wells, G. (2006). Shelter-based managed alcohol administration to chronically homeless people addicted to alcohol. *Canadian Medical Association Journal*, 174(1):45–49. Documents the Ottawa Managed Alcohol Program; controlled alcohol administration in a supervised shelter setting produced measurable reductions in alcohol consumption, emergency department visits, and police encounters.
- Substance Abuse and Mental Health Services Administration (2024). Assertive community treatment evidence-based practices implementation resource kit. Technical report, SAMHSA, U.S. Department of Health and Human Services. Documents the federal ACT implementation grant program; competitive grants of five hundred thousand to two million dollars per program; California historically a strong SAMHSA ACT grantee.
- UCSF Benioff Homelessness and Housing Initiative (2023). Toward a new understanding: The California statewide study of people experiencing homelessness. Technical report, UCSF Benioff Homelessness and Housing Initiative. Documents the sub-forty percent housing retention outcomes for highly vulnerable cohorts absent continuous clinical stabilization prerequisites.
- United States Congress (1933). Tennessee valley authority act of 1933. 16 U.S.C. §§ 831-831ee. Establishes the foundational federal precedent for a quasi-governmental, state-chartered stewardship authority wielding delegated police power and independent bonding capacity for public infrastructure operations.
- United States Congress (1972). Federal water pollution control act (Clean Water Act), 33 U.S.C. § 1251 et seq. Public Law 92-500. Section 402 establishes the NPDES permit system governing point source discharges into navigable waters; provides the federal regulatory basis for environmental enforcement against riparian encampment contamination.
- United States Department of Agriculture (2023). Animal companions and shelter refusal patterns among the unsheltered population. Technical report, USDA, Animal and Plant Health Inspection Service. Identifies pet separation as a primary stated reason for shelter refusal; documents the role of animal companions as emotional regulation and survival anchors for unsheltered individuals.
- University of Southern California (2024). Digital surveillance and ontological security in affordable housing. *USC Urban Policy Research*. Documents the inverse relationship between in-unit digital monitoring density and resident-reported ontological security in multi-unit affordable housing complexes.
- U.S. Department of Housing and Urban Development (2022). Barriers to shelter utilization among the unsheltered homeless population. Technical report, HUD Office of Policy Development and Research. Documents the Three Ps refusal pattern (pets, partners, possessions) as the primary driver of voluntary shelter non-participation among the encamped engageable population.

World Health Organization (2006). Oral rehydration salts: Production of the new ORS. *WHO Technical Document*. WHO-formulated oral rehydration solution specification; establishes the clinical standard for electrolyte-balanced rehydration preventing hyponatremia in severely dehydrated individuals.

Y-Foundation (2023). A home of your own: Housing first and ending homelessness in Finland. Technical report, Y-Säätiö (Y-Foundation), Helsinki. Documents sixty-eight percent reduction in long-term homelessness in Finland 2008–2022; details the Housing First national program, institutional landlord model, and cost-effectiveness evidence.